

# Deixis

*Typeset decoupled notes, connectors, and spatial highlights.*

Version 0.1.0

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# 1. Introduction

deixis is a unified layout engine for inline notes, footnotes, endnotes, margin notes, inset notes, and inline spatial highlights with visual connectors.

## 1.1. Quick Start

To use deixis, you must initialize the state engine at the top of your document:

```
#import "@preview/deixis:0.1.0": *  
#show: deixis-setup-notes
```

## 1.2. Core Concepts

Inspired by `\footnotemark` and `\footnotetext` in LaTeX, deixis decouples an annotation into distinct components:

- **The mark:** The visual indicator placed directly within your text flow. It increments counters and records its exact physical coordinates in the document.

```
#deixis-inline-mark(  
  id: <anatomy>,  
) [*Note mark*]
```

Note mark<sup>1</sup>

- **The body:** The actual content of your note. It can be rendered anywhere – floating in the margin, at the bottom of the page, or at the end of a chapter – while maintaining a bi-directional hyperlink to the mark if they share the same id.

```
#deixis-inset-note-body(  
  id: <anatomy>,  
  link: "right-angle",  
  link-waypoints: ((200pt, "body")),  
  link-marks: "body",  
  label: <anatomy>,  
  backlink: true,  
) [*Note body.*]
```

<sup>1</sup>Note body.  

- **The link:** The optional connection that links the mark to the body. Semantically, it can be considered the third component, but programatically, it is treated as part of the body (set via `link`).

If given a `label`, notes can be referenced anywhere in the document. The body is also aware of where it is referenced, allowing it to generate backlink buttons .

See @anatomy.

See <sup>1</sup>.



“Still unamused? There are way more to discover. 😊”

## 2. API Reference

### 2.1. Setup & Configuration

#### `deixis-setup-notes()`

Initializes the deixis state engine and layout overlays.

This function **must** wrap your document `#show: deixis-setup-notes.with(...)` before any other deixis functionality can be used.

#### ⚠ Warning

deixis uses the page foreground and background to display some notes. Any attempt to modify the foreground or background after `#show: deixis-setup-notes` will completely overwrite the setup. Thus, you must always put `#set page(background: ..., foreground: ...)` before `#show: deixis-setup-notes`.

#### Parameters

```
deixis-setup-notes(  
  body: content,  
  foreground: none or content,  
  background: none or content,  
  override-page-footnotes: bool  
) -> content
```

**body**     `content`

The main document content to wrap.

**foreground**     `none` or `content`

Custom content to inject into the foreground overlay layer.

Default: `none`

**background**     `none` or `content`

Custom content to inject into the background overlay layer.

Default: `none`

**override-page-footnotes**     `bool`

If `true`, page-level footnotes will be rendered on the foreground.

#### **i** Info

Foreground overlay allows some behaviors like grouping footnotes by series or aligned marker.

Default: `true`

## deixis-set()

Dynamically updates the global deixis configuration mid-document.

This function acts like a standard Typst `#set` rule for the internal deixis state. It applies your new configuration to all subsequent notes within the current scope.



### Tip

**Type-Specific Styling:** All parameters accept either a global value or a dictionary targeting specific note types (e.g., `stroke: (margin-note: red, inset-note: blue)`). The dictionary will be interpreted as scope specific and not a global value if it contains either note-scope keys or component-scope keys.

- Note-scope keys: (`"inline-mark"`, `"phantom-mark"`, `"region-mark"`, `"footnote"`, `"endnote"`, `"margin-note"`, `"sidenote"`, `"inset-note"`, `"inline-note"`, `)`.
- Component-scope keys: (`"mark"`, `"body"`, `"link"`).
  - `"nodes"` indicates both `"mark"` and `"body"` (excluding `"link"`).
- Wildcard key: `"rest"` indicates anything else.

## Parameters

```
deixis-set(  
  numbering: auto str function dictionary ,  
  marker-style: auto function dictionary ,  
  body-style: auto function dictionary ,  
  backlink: auto bool str dictionary ,  
  marker-position: auto alignment dictionary ,  
  inline-mode: auto str ,  
  stroke: auto stroke none dictionary ,  
  fill: auto color none dictionary ,  
  radius: auto length dictionary ,  
  link: auto str dictionary ,  
  link-marks: auto str dictionary ,  
  separator: auto content none dictionary ,  
  clearance: auto length dictionary ,  
  gap: auto length dictionary ,  
  marker-gap: auto length dictionary ,  
  indent: auto length dictionary ,  
  margin-layout: auto str ,  
  min-margin-width: auto length ,  
  side-strategy: auto str ,  
  mark-align: auto alignment dictionary ,  
  mark-align-strictness: auto str ,  
  spillover: auto bool ,  
  region-shape: auto function ,  
  layer: auto str ,  
  render-single: auto function dictionary ,  
  render-group: auto function dictionary ,  
  container-func: auto function dictionary  
) -> content
```

**numbering** auto or str or function or dictionary

The numbering style for the marker.

```
#deixis-set(numbering: "i")  
#deixis-footnote[A note with roman  
numbering.]
```

i

---

<sup>i</sup>A note with roman numbering.

Default: auto

**marker-style** auto or function or dictionary

The styling function (content) => content for note markers.

```
#deixis-set(marker-style: (  
  mark: it => text(super(it), fill: red),  
  body: it => text([#it.], fill: blue,  
  weight: "bold"))  
)  
#deixis-footnote[A note with custom mark  
marker style and body marker style.]
```

1

---

**1.**A note with custom mark marker style and body marker style.

Default: **auto**

**body-style** **auto** or **function** or **dictionary**

The styling function (content) => content for note bodies.

```
#deixis-set(body-style: it => text(it,
fill: blue, weight: "semibold"))
#deixis-footnote[A note with custom body
style.]
```

<sup>1</sup>  
A note with custom body style.

### **i** Info

Setting body-style will override the default styling, i.e. it => text(size: 0.85em, it). Instead, applying custom styling directly to the body content does not.

```
#deixis-footnote[#text(fill: blue,
weight: "semibold")][Another note with
custom body style.]
```

<sup>1</sup>  
Another note with custom body style.

Default: **auto**

**backlink** **auto** or **bool** or **str** or **dictionary**

Enables or disables clickable backlinks from bodies to marks. Choices: **bool** | "always" | "none" | "never" | "multiple".

```
#deixis-set(backlink: true)
#deixis-footnote(label: <deixis-
set.backlink>)[A note with backlink.]
#deixis-ref(<deixis-set.backlink>)
```

<sup>1</sup> <sup>1</sup>  
A note with backlink. <sup>1</sup> <sup>2</sup>

Default: **auto**

**marker-position** **auto** or **alignment** or **dictionary**

Mark marker placement.

See deixis-inline-mark.marker-position and deixis-region-mark.marker-position.

Default: **auto**

**inline-mode** **auto** or **str**

The visual decoration applied to the marked inline text. Choices: "highlight" | "box" | "parentheses" | "underline" | "text-fill" | "none".

See deixis-inline-mark.inline-mode.

Default: **auto**

**stroke** auto or stroke or none or dictionary

The border stroke.

```
#deixis-inset-note(  
  stroke: (mark: red, body: blue),  
  layer: "flow",  
) [A note with red mark stroke.] [And blue  
body stroke.]
```

A note with red mark stroke.<sup>1</sup>

<sup>1</sup>And blue body stroke.

Default: auto

**fill** auto or color or none or dictionary

The background fill color.

```
#deixis-inset-note(  
  stroke: none,  
  fill: (mark: red, body: blue),  
  layer: "flow",  
) [A note with red mark fill.] [And blue  
body fill.]
```

A note with red mark fill.<sup>1</sup>

<sup>1</sup>And blue body fill.

Default: auto

**radius** auto or length or dictionary

The border radius.

```
#deixis-inset-note(  
  stroke: yellow,  
  fill: yellow.transparentize(95%),  
  radius: (mark: 0pt, body: 1em),  
  layer: "flow",  
) [A note with `0pt` mark radius.] [And  
`1em` body radius.]
```

A note with 0pt mark radius.<sup>1</sup>

<sup>1</sup>And 1em body radius.

Default: auto

**link** auto or str or dictionary

The type of connector line. Choices: "none" | "straight-line" | "right-angle" | "chamfer" | "curve" | "ucr" | "ccr".

- "none": The link is completely invisible.
- "straight-line": A direct, shortest-path line from the start to the end point.

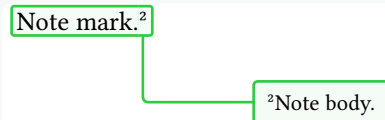
```
#deixis-inset-note(  
  stroke: green,  
  fill: green.transparentize(95%),  
  link: "straight-line",  
  dx: 4em, dy: 3em,  
  layer: "flow",  
) [Note mark.] [Note body.]
```

Note mark.<sup>1</sup>

<sup>1</sup>Note body.

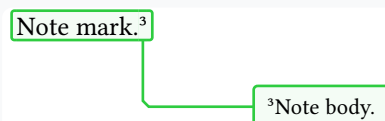
- **"right-angle"**: Routes the link orthogonally (moving only horizontally and vertically) and smooths the turns with rounded corners (fillets).

```
#deixis-inset-note(
  stroke: green,
  fill: green.transparentize(95%),
  link: "right-angle",
  dx: 4em, dy: 3em,
  layer: "flow",
)[Note mark.][Note body.]
```



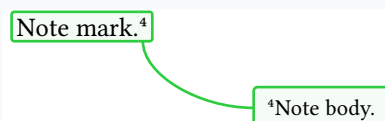
- **"chamfer"**: Similar to **"right-angle"**, but cuts the corners with diagonal straight lines (bevels) instead of rounding them.

```
#deixis-inset-note(
  stroke: green,
  fill: green.transparentize(95%),
  link: "chamfer",
  dx: 4em, dy: 3em,
  layer: "flow",
)[Note mark.][Note body.]
```



- **"curve"**: Draws a smooth **Weighted Bessel Spline**.

```
#deixis-inset-note(
  stroke: green,
  fill: green.transparentize(95%),
  link: "curve",
  dx: 4em, dy: 3em,
  layer: "flow",
)[Note mark.][Note body.]
```



- **"ccr"**: Draws a **Centripetal Catmull-Rom** spline, which is more stable compared to other curves and less prone to unnatural loops or aggressive bulges.
- **"ucr"**: Draws a standard **Uniform Catmull-Rom** spline.

### 🔥 Tip

Link type can be changed dynamically mid-path by declaring them in the `link-waypoints` parameters of the note functions. See `deixis-common-note-args.link-waypoints`.

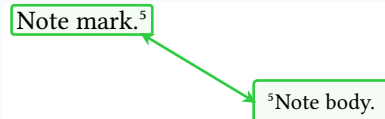
Default: **auto**

**link-marks** **auto** or **str** or dictionary

The connector arrowhead style. Choices: **"none"** | **"mark"** | **"body"** | **"both"**.



```
#deixis-inset-note(
  stroke: green,
  fill: green.transparentize(95%),
  link: "straight-line",
  link-marks: "both",
  dx: 4em, dy: 3em,
  layer: "flow",
)[Note mark.][Note body.]
```



Default: **auto**

**separator** **auto** or **content** or **none** or dictionary

The separator line for note groups.

### **i** Info

This only has effect on grouped notes (footnote and endnote).

```
#deixis-set(separator: line(length: 50%,
  stroke: (thickness: 0.5pt, dash:
    "dashed")))
#deixis-footnote[A footnote separated by
  a dashed line.]
```

1  
-----  
<sup>1</sup>A footnote separated by a dashed line.

See also `std.footnote.entry.separator`.

Default: **auto**

**clearance** **auto** or **length** or dictionary

Vertical spacing before note groups.

### **i** Info

This only has effect on grouped notes (footnote and endnote).

```
#deixis-set(clearance: 3em)
#deixis-footnote[A note `3em`-spaced from
  the main content.]
```

1  
  
-----  
<sup>1</sup>A note 3em-spaced from the main content.

See also `std.footnote.entry.clearance`.

Default: **auto**

**gap** **auto** or **length** or dictionary

Vertical spacing between adjacent notes.

## **i** Info

This only has effect on grouped notes (footnote and endnote) and margin note.

```
#deixis-set(gap: 1em)
#deixis-footnote[Note A.]
#deixis-footnote[Note B.]
```

12

---

<sup>1</sup>Note A.  
<sup>2</sup>Note B.

See also `std.footnote.entry.gap`.

Default: **auto**

**marker-gap** **auto** or **length** or **dictionary**

Horizontal spacing between the marker and the body text.

```
#deixis-set(marker-gap: 1em)
#deixis-footnote[A note with `marker-
gap=1em`.]
```

1

---

<sup>1</sup> A note with marker-gap=1em.

Default: **auto**

**indent** **auto** or **length** or **dictionary**

The indentation applied to the note body.

```
#deixis-set(indent: 3em)
#deixis-footnote[A note with
`indent=3em`.]
```

1

---

<sup>1</sup>A note with indent=3em.

See also `std.footnote.entry.indent`.

Default: **auto**

**margin-layout** **auto** or **str**

The layout engine for margin notes. Choices: **"adaptive"** | **"flow"** | **"exact"**.

- **"adaptive"**: Apply a two-pass *rubber band algorithm* to calculate the y-coordinates of all margin notes in the page; then use `#std.place` to put them into calculated positions.
- **"flow"**: Put margin notes in a block container with interleaved `#v(space)`, where `space` is provisional inter-note gap converted to **fraction**; and let the Typst engine calculates their final positions.
- **"exact"**: Put margin notes exactly where the marker is, ignore if they overlap or not.

Default: **auto**

**min-margin-width** **auto** or **length**

Minimum margin width to put margin notes.

```
The margin of the minipage is set to  
(left: 1.2in, right: 0.5in).  
#deixis-set(min-margin-width: 1in)  
#deixis-margin-note[This note prefers the  
left margin.]
```

```
The margin of the  
minipage is set to  
^This note prefers (left: 1.2in,  
the left margin. right: 0.5in).^
```

Default: **auto**

**side-strategy** **auto** or **str**

Strategy to select side for margin note. Choices: "nearest" | "strict".

See `deixis-margin-note-body.side-strategy`.

Default: **auto**

**mark-align** **auto** or **alignment** or **dictionary**

How margin notes align vertically relative to their marks.

See `deixis-margin-note-body.mark-align`.

Default: **auto**

**mark-align-strictness** **auto** or **str**

How strictly to align margin notes to their marks. Choices: "strict" | "loose" | "none".

See `deixis-margin-note-body.mark-align-strictness`.

Default: **auto**

**spillover** **auto** or **bool**

Allow margin notes to spill over page boundaries.

See `deixis-margin-note-body.spillover`.

Default: **auto**

**region-shape** **auto** or **function**

Shape drawing function for region marks.

See `deixis-region-mark.region-shape`.

Default: **auto**

**layer** **auto** or **str**

The rendering layer for region marks and inset notes. Choices: "flow" | "foreground" | "background".

See `deixis-region-mark.layer`.

Default: **auto**

**render-single** **auto** or **function** or **dictionary**

Note body rendering function.

render-single is used in combination with render-group. deaxis provides a few basic render functions:

- **native**: *Almost* pixel-identical to std. footnote.

```
#deaxis-set(  
  render-single: deaxis-native-render-  
    single,  
  render-group: deaxis-native-render-  
    group,  
)  
#deaxis-footnote[A normal note.]  
#deaxis-footnote(marker: [long marker])[A  
note with very long marker, and also a  
long body.]
```

1long marker

---

<sup>1</sup>A normal note.  
long marker A note with very long marker, and also a  
long body.

- **default**: Body markers are aligned, but the notes are still prose-printed.

```
#deaxis-set(  
  render-single: deaxis-default-render-  
    single,  
  render-group: deaxis-default-render-  
    group,  
)  
#deaxis-footnote[A normal note.]  
#deaxis-footnote(marker: [long marker])[A  
note with very long marker, and also a  
long body.]
```

1long marker

---

<sup>1</sup>A normal note.  
long marker A note with very long marker, and also a  
long body.

- **grid**: Body markers and body content are structured in a grid.

```
#deaxis-set(  
  render-group: deaxis-grid-render-group,  
)  
#deaxis-footnote[A normal note.]  
#deaxis-footnote(marker: [long marker])[A  
note with very long marker, and also a  
long body.]
```

1long marker

---

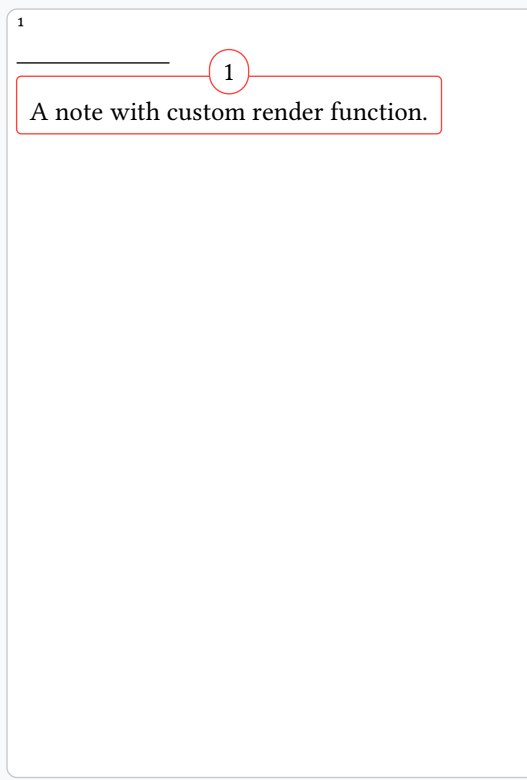
|              |                                                        |
|--------------|--------------------------------------------------------|
| <sup>1</sup> | A normal note.                                         |
| long marker  | A note with very long marker, and also a<br>long body. |

### Tip

For full control, you can (and should) draft your own render function. deaxis provides some handy helper functions for this purpose. See `deaxis-generate-backlinks()`, `deaxis-generate-body-meta()`, and `deaxis-generate-body-marker()`.

```
#let custom-render-func(note-
data, ..args) = {
  let backlink = note-data.backlink
  let body-marker = deixis-generate-body-
marker(note-data + (body-marker-style: it
=> it))
  rect(
    ..note-data.styles,
    inset: (top: 10pt, rest: 5pt)
  ) [
    #place(center + top, dy: -20pt, rect(
      stroke: note-data.styles.stroke,
      fill: white,
      radius: 1em,
      body-marker))
    #note-data.body
  ]
}

#deixis-set(
  render-single: custom-render-func,
  render-group: deixis-native-render-
group,
)
#deixis-footnote[A note with custom
render function.]
```



Default: **auto**

**render-group** **auto** or **function** or **dictionary**

Group rendering function for grouped notes (footnote and endnote).

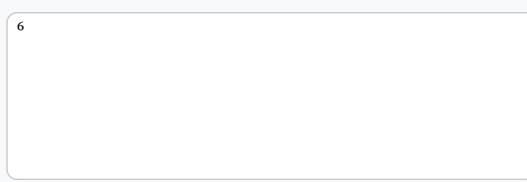
See `deixis-set.render-single`.

Default: **auto**

**container-func** **auto** or **function** or **dictionary**

Note body container function with signature `(content, ..args) => content`.

```
#import "@preview/colorful-boxes:1.4.3":
stickybox
#deixis-set(container-func:
  (body, ..args) => stickybox(body))
#deixis-margin-note(target: "page") [Yes,
we can also do that.]
```



“Yes, we can  
also do that.”

### 🔥 Tip

`container-func` is only a wrapper around the rendered body of each note and does not have access to the raw notes’ data. For more sophisticated control, you should use `deixis-set.render-single` or `deixis-set.render-group`.

Default: **auto**

## 2.2. Common Mechanics

`deixis-common-note-args()`

### ⚠ Warning

This is not a real function, added only for documentation purposes.

This is a list of common arguments of note functions in the `deixis` package and their usage.

#### Parameters

```
deixis-common-note-args(  
  mark-type: auto str,  
  id: none str label,  
  numbering: auto str function,  
  target: auto int label str,  
  series: str,  
  marker: auto content,  
  stroke: auto stroke none,  
  fill: auto color none,  
  radius: auto length dictionary,  
  link: auto str,  
  link-waypoints: none array,  
  link-ports: auto array dictionary,  
  link-marks: auto str,  
  render-single: auto function,  
  container-func: auto function,  
  ..args: arguments  
) -> content
```

**mark-type** auto or str

The type of mark to render in the text. Choices: "inline" | "region" | "phantom".

This argument is present in all wrapper note functions `deixis-footnote()`, `deixis-endnote()`, `deixis-margin-note()`, and `deixis-inset-note()`. These note functions create both the mark and the body, and oftentimes it is ambiguous to infer which mark function to use internally. `mark-type` can then be used to specify this.

```
#deixis-margin-note(  
  mark-type: "region",  
  padding: "text",  
  target: "page",  
)[$ e^(i pi) + 1 = 0 $][Euler's  
Identity.]
```

$$e^{i\pi} + 1 = 0^7$$

<sup>7</sup>Euler's Identity.

### 🔥 Tip

The allowed choices also include verbose names: "inline-mark" | "region-mark" | "phantom-mark".

Default: auto

**id** none or str or label

A unique identifier for linking a mark to a decoupled body.

```
#deixis-inline-mark(id: <common-note.id>)[This is a standalone inline note.]  
#deixis-margin-note-body(id: <common-note.id>, target: "page")[This is a decoupled margin note body linked to the standalone inline note.]
```

This is a standalone inline note.<sup>1</sup>

<sup>1</sup>This is a decoupled margin note body linked to the standalone inline note.

### **i** Info

For celibate marks, if an id is given but the mark is linked to no body, the mark will be numbered with a marker but not clickable.

Default: none

**numbering** auto or str or function

The numbering style (used for the body and outline, though invisible here).

```
#deixis-footnote(numbering: "i")[A note with roman numbering.]  
#deixis-footnote(numbering: "*")[A note with symbol numbering.]  
#deixis-footnote(numbering: it => std.numbering("1.1"), counter(heading).get().first(), it))[A note with functional numbering.]
```

i†(2.3)

<sup>i</sup>A note with roman numbering.  
<sup>†</sup>A note with symbol numbering.  
<sup>(2.3)</sup>A note with functional numbering.

Default: auto

**target** auto or int or label or str

The specific block/minipage counter context a mark belongs to, or a specific render context a note body belongs to.

- If "page", the note will be a page-level note regardless of where it is.

```
#deixis-footnote(target: "page")[This is a page-level footnote.]  
#deixis-margin-note(target: "page")[This is a page-level margin note.]
```

89

<sup>9</sup>This is a page-level margin note.

- If a label id of a `deixis-block()` is provided, the note will be rendered as part of that block. This is especially useful for footnote and margin note.

<sup>8</sup>This is a page-level footnote.

```
#deixis-footnote(target: <common-note.target>)[This footnote is targeted
at the gray block.]
#deixis-block(
  id: <common-note.target>,
  fill: gray.lighten(50%),
  inset: 0.1em)
External notes can target a block
*regardless* of their relative position
being before or after the block.
#deixis-footnote(target: <common-note.target>)[And so is this footnote.]
```

1

<sup>1</sup>This footnote is targeted at the gray block.  
<sup>2</sup>And so is this footnote.

External notes can target a block **regardless** of their relative position being before or after the block.<sup>2</sup>

- If a **non-positive** `int` is provided, the target will be resolved based on the current stack level. For example, value of `0` means this current block/minipage, a value of `-1` means the parent block/minipage, and so on.
- An `auto` is interpreted as `0` in mark functions and to *inherit* the target of the mark in body functions.



### Tip

- For mark-only note functions (e.g. `deixis-inline-mark()` and `deixis-region-mark()`), target decides which counter system the note uses. The mark is always placed where it is called.
- For body-only note functions (e.g. `deixis-margin-note-body()` or `deixis-inset-note-body()`), target decides where to render the body.

Default: `0`

**series** `str`

The counter series this note belongs to.

For grouped notes like footnote, notes from the same series are also grouped together.

```
This is a translation note#deixis-footnote(series: "trans", numbering: "*")
[Old Latin: 'Salvé'].
This is a general comment#deixis-footnote(series: "comm", numbering: "1")
[Added by the editor.].
Another translation note#deixis-footnote(series: "trans", numbering: "*")
[Modern: 'Hello'].
And another comment#deixis-footnote(series: "comm", numbering: "1")
[Added by the reviewer.].
```

This is a translation note<sup>\*</sup>. This is a general comment<sup>†</sup>. Another translation note<sup>†</sup>. And another comment<sup>2</sup>.

<sup>\*</sup>Old Latin: 'Salvé'.

<sup>†</sup>Modern: 'Hello'.

<sup>1</sup>Added by the editor.

<sup>2</sup>Added by the reviewer.

Default: `"default"`

**marker** `auto` or `content`

A hardcoded marker override.



One can also use numbering: it => content to replace the marker with any content they want, marker is just a short-hand syntactic sugar.

```
#deixis-footnote(marker: emoji.gear)[A  
note with _gear_ marker.]
```



A note with *gear* marker.

Default: **auto**

**stroke** **auto** or **stroke** or **none**

The border stroke applied to the mark, the note body, and/or the link.

Default: **auto**

**fill** **auto** or **color** or **none**

The background fill color applied to the mark and/or the note body.

Default: **auto**

**radius** **auto** or **length** or **dictionary**

The border radius applied to the mark, the note body, and/or the link.

Default: **auto**

**link** **auto** or **str**

The type of connector line. Choices: **"none"** | **"straight-line"** | **"right-angle"** | **"chamfer"** | **"curve"** | **"ucr"** | **"ccr"**.

See `deixis-set.link`.

Default: **auto**

**link-waypoints** **none** or **array**

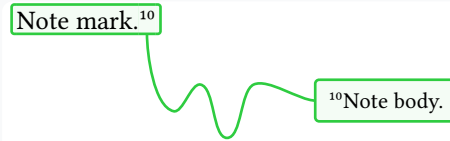
An array of intermediate coordinate offsets to route the connector line through.

Waypoints allow you to manually detour the link around obstacles or create custom geometric paths. The array can mix several types of instructions to build complex paths:

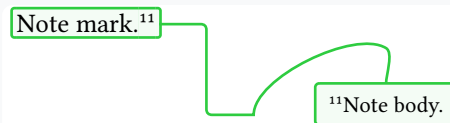
- Relative coordinates ( **array** ): Offsets relative to the previous point in the path.
- Component keywords (**"mark"** | **"body"**): offsets to the respective component's absolute coordinates.
  - ▶ Component keywords can be mixed with **alignment** keywords like **"body-top-left"** to offset to a corner of that component.
- Link types (**"none"** | **"straight-line"** | **"right-angle"** | **"curve"** | **"ucr"** | **"ccr"**): Changes link type mid-path.
- **"split"**: A reserved keyword for splitting the paths of spoiled notes, which have marks and bodys lying in different pages. It can also be put in a tuple:
  - ▶ (**"split"**, **"mark"**): Spills at the x-coordinate of the mark.

- ▶ ("split", "body"): Spills at the x-coordinate of the body.
- ▶ ("split", "center"): Spills at the center x-coordinates of the mark and the body.
- ▶ ("split", 10pt): Spills at a x-offset relative to the previous position in the path.

```
#deixis-inset-note(
  stroke: green,
  fill: green.transparentize(95%),
  link: "curve",
  link-waypoints: ((10pt, 40pt), (10pt,
-10pt), (10pt, 20pt), (10pt, -20pt)),
  dx: 6em, dy: 3em,
  layer: "flow",
)[Note mark.][Note body.]
```



```
#deixis-inset-note(
  stroke: green,
  fill: green.transparentize(95%),
  link: "right-angle",
  link-waypoints: ((40pt, 40pt), "curve",
("body-top-center", -25pt)),
  dx: 6em, dy: 3em,
  layer: "flow",
)[Note mark.][Note body.]
```

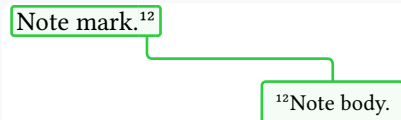


Default: none

**link-ports** **auto** or array or dictionary

Defines the exact exit/entry ports for the connector line.

```
#deixis-inset-note(
  stroke: green,
  fill: green.transparentize(95%),
  link: "right-angle",
  link-ports: (mark: bottom, body: top),
  dx: 4em, dy: 3em,
  layer: "flow",
)[Note mark.][Note body.]
```



## i Info

The available ports for `deixis-region-mark()` and any note body are:

- left | right | top | bottom.

The available ports for `deixis-inline-mark()` are:

- If `marker-position: right`: right | top | bottom.
- If `marker-position: left`: left | top | bottom.

If **auto**, the ports are automatically selected based on the relative positions of the mark and the body, or based on the first & last waypoint coordinates if `link-waypoints` is provided.

Default: auto

**link-marks** `auto` or `str`

The arrowhead style for the connector line. Choices: `"none"` | `"mark"` | `"body"` | `"both"`.

See `deixis-set.link-marks`.

Default: `auto`

**render-single** `auto` or `function`

The width of the inset note body. -> `auto` | `length` | `relative` A custom render function for the inner layout of the body.

See `deixis-set.render-single`.

Default: `auto`

**container-func** `auto` or `function`

A custom container wrapper for the body block.

See `deixis-set.container-func`.

Default: `auto`

**..args** `arguments`

Positional arguments are parsed in the following order: `[mark][body]`.

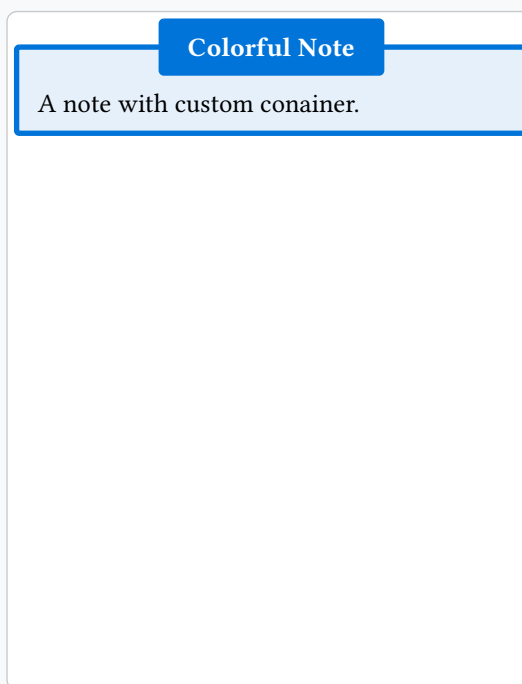
- `[mark]`: the marked content to be highlighted.
- `[body]`: the content of the note body.

All named arguments are forwarded along with `stroke`, `fill`, `radius` as styles to the container.

```
#import "@preview/colorful-boxes:1.4.3":
outline-colorbox

#let colorbox-container(body, title:
none, stroke: none, fill: none, radius:
0pt, centering: false, ..args) = outline-
colorbox(
  title: title,
  color: (fill: fill, stroke:
stroke.paint),
  radius: radius,
  centering: centering,
)[#body]

#deixis-inline-note-body(
  container-func: colorbox-container,
  // all these parameters are passed to
colorbox-container
  stroke: blue,
  fill: blue.transparentize(90%),
  title: [Colorful Note],
  centering: true
)[A note with custom conainer.]
```



## 2.3. Marks

### 2.3.1. Inline Mark

#### `deixis-inline-mark()`

An inline text mark with a visual marker.

This function drops a numbered (or custom) marker into the text and registers its location in the state engine.

Inline note is usually just a single marker `#deixis-inline-mark(id: <celibate-inline-note1>)`.  
Alternatively, it can wrap `#deixis-inline-mark(id: <celibate-inline-note2>)[a text content of arbitrary length with it]`.

Inline note is usually just a single marker<sup>1</sup>.  
Alternatively, it can wrap a text content of arbitrary length with it<sup>2</sup>.

#### Parameters

```
deixis-inline-mark(  
  ..args: arguments ,  
  numbering: auto str function ,  
  marker: auto content ,  
  target: int label str ,  
  series: str ,  
  id: none str label ,  
  marker-style: auto function ,  
  marker-position: auto alignment ,  
  inline-mode: auto str ,  
  stroke: auto stroke none ,  
  fill: auto color none ,  
  radius: auto length dictionary  
) -> content
```

**..args** arguments

Accepts up to one positional arguments: [body].

**numbering** auto or str or function

The numbering style (e.g., "1", "a") for the generated marker.

Default: auto

**marker** auto or content

A hardcoded marker (like a symbol) to use instead of the automatic counter.

Default: auto

**target** int or label or str

The specific block/minipage counter context this mark belongs to.

Default: 0

**series** `str`

The counter series this mark belongs to.

Default: `"default"`

**id** `none` or `str` or `label`

A unique identifier linking this mark to its corresponding body.

Default: `none`

**marker-style** `auto` or `function`

A custom styling function (string) => content for the generated marker.

Default: `auto`

**marker-position** `auto` or `alignment`

The alignment/positioning of the marker relative to its anchor.

```
#deixis-inline-mark(marker-position:
left, id: <celibate-inline-note3>)[
  This noted content has its mark placed
  on the left side.
]
```

This noted content has its mark placed on the left side.

Default: `auto`

**inline-mode** `auto` or `str`

The visual decoration applied to the marked inline text. Choices: `"highlight"` | `"box"` | `"parentheses"` | `"underline"` | `"text-fill"` | `"none"`.

```
#deixis-inline-mark(inline-mode:
"highlight")[Marked content]
```

Marked content

```
#deixis-inline-mark(inline-mode: "box")
[Marked content]
```

Marked content

```
#deixis-inline-mark(inline-mode:
"parentheses")[Marked content]
```

[Marked content]

```
#deixis-inline-mark(inline-mode:
"underline")[Marked content]
```

Marked content

```
#deixis-inline-mark(inline-mode: "text-
fill")[Marked content]
```

Marked content

```
#deixis-inline-mark(inline-mode: "none")  
[Marked content]
```

Marked content

Default: **auto**

**stroke** **auto** or **stroke** or **none**

The border stroke.

Default: **auto**

**fill** **auto** or **color** or **none**

The background fill color.

Default: **auto**

**radius** **auto** or **length** or **dictionary**

The border radius.

Default: **auto**

### 2.3.2. Phantom Mark

#### `deixis-phantom-mark()`

A semantic mark that is registered without leaving *any* visible trace in the text.

Phantom marks do increment the counter and register their position in the state engine, but leave no visual footprint in the paragraph. This is useful for creating semantic anchors or side-channel annotations that shouldn't disrupt the reading flow.

```
The library was constructed with massive
stone walls to regulate the indoor climate
naturally across the changing seasons.
#deixis-phantom-mark(id: <phantom>, marker:
none)
This architectural choice ensured that
centuries of fragile manuscripts remained
perfectly preserved without the need for
artificial climate control.

#deixis-margin-note-body(id: <phantom>)
[Thick stone absorbs daytime heat and
releases it at night.]
```

The library was constructed with massive stone walls to regulate the indoor climate naturally across the changing seasons. This architectural choice ensured that centuries of fragile manuscripts remained perfectly preserved without the need for artificial climate control.

Thick stone absorbs daytime heat and releases it at night.

#### Parameters

```
deixis-phantom-mark(
  id: none or str or label,
  numbering: auto or str or function,
  target: int or label or str,
  series: str,
  marker: auto or content
) -> content
```

**id** none or str or label

A unique identifier for linking this mark to a decoupled body.

Default: `none`

**numbering** auto or str or function

The numbering style (used for the body and outline, though invisible here).

Default: `auto`

**target** int or label or str

The specific block/minipage counter context this mark belongs to.

Default: `0`

**series** str

The counter series this mark belongs to.

Default: `"default"`

**marker** auto or content

A hardcoded marker override (used for the body and outline, though invisible here).

Default: **auto**



### 2.3.3. Region Mark

#### deixis-region-mark()

A visual enclose mark of a specific region in the document.

Region marks are highly versatile. They can dynamically wrap a continuous block of text provided as a positional argument, or they can asynchronously draw a shape spanning between pre-defined #deixis-pin coordinates. They natively support crossing page boundaries when using the background or foreground layers.

Inline marks cannot highlight equations or raw:

```
#deixis-inline-mark[*Sigmoid:* $f(x) =  
frac(1, 1 + e^(-x))$ or `torch.sigmoid`.]
```

This is where region marks shine:

```
#deixis-region-mark(inline: true)[*ReLU:*  
$f(x) = max(0, x)$ or `torch.relu`]
```

Inline marks cannot highlight equations or raw:

**Sigmoid:**  $f(x) = \frac{1}{1+e^{-x}}$  or torch.sigmoid.

This is where region marks shine:

**ReLU:**  $f(x) = \max(0, x)$  or torch.relu

```
#let highlight-cell = deixis-region-  
mark.with(stroke: blue, fill:  
blue.transparentize(95%))
```

```
#set text(0.8em)
```

```
#table(  
  columns: (auto, 1fr, auto, 1fr),  
  align: center + horizon,  
  fill: (x, y) => if y == 0  
{ gray.lighten(80%) },
```

```
  table.header([*Function*], [*Equation*],  
    [*Output\ Range*], [*Common\ Usecase*]),  
    [*Sigmoid*], [$1 / (1 + e^(-x))$], [$(0,  
1)$], [Binary classifier],  
    [*Tanh*], [$(e^x - e^(-x)) / (e^x + e^(-  
x))$], [$( -1, 1)$], [RNNs],  
    [*ReLU*], [$(max(0, x))$], [$(0, inf)$],  
    [CNNs],  
    [*GELU*], highlight-cell(padding: 4pt)[$x  
Phi(x)$], [$(approx [-0.17, inf))$],  
    [*Transformers*],  
)
```

| Function | Equation                            | Output Range                  | Common Usecase    |
|----------|-------------------------------------|-------------------------------|-------------------|
| Sigmoid  | $\frac{1}{1+e^{-x}}$                | (0, 1)                        | Binary classifier |
| Tanh     | $\frac{e^x - e^{-x}}{e^x + e^{-x}}$ | (-1, 1)                       | RNNs              |
| ReLU     | $\max(0, x)$                        | [0, inf)                      | CNNs              |
| GELU     | $x\Phi(x)$                          | $\approx [-0.17, \text{inf})$ | Transformers      |

## Parameters

```
deixis-region-mark(  
  ..args: arguments ,  
  id: none str label ,  
  pins: array ,  
  numbering: auto str function ,  
  marker: auto content ,  
  target: int label str ,  
  series: str ,  
  marker-style: auto function ,  
  marker-position: auto alignment dictionary ,  
  stroke: auto stroke none ,  
  fill: auto color none ,  
  radius: auto length dictionary ,  
  padding: auto length str ,  
  region-shape: auto function ,  
  layer: auto str ,  
  inline: bool  
) -> content
```

**..args** arguments

Accepts at most 1 positional argument: [body].

**id** none or str or label

The unique identifier linking this region to its data/body state.

Default: none

**pins** array

An array of pin names (e.g., ("pinA", "pinB")). If provided, the note draws between these pins instead of wrapping a positional body. Cannot be used simultaneously with a positional body argument.

See `deixis-pin()` and `deixis-attach()`.

Default: ()

**numbering** auto or str or function

The numbering style for the region's marker.

Default: auto

**marker** auto or content

A hardcoded marker to use instead of the automatic counter.

Default: auto

**target** int or label or str

The specific block/minipage counter context this mark belongs to.

Default: 0

**series** str

The counter series this mark belongs to.

Default: "default"

**marker-style** auto or function

A custom styling function (string) => content for the generated marker.

Default: auto

**marker-position** auto or alignment or dictionary

Where to place the marker relative to the bounding box.

- If alignment, align to one of the 9 locations.

```
#deixis-region-mark(  
  id: <celibate-region1>,  
  marker-position: top + center)  
[#lorem(10)]
```

- If dictionary of length or relative, place the mark at (dx, dy) relative to the top left corner of the region. relative is understood as to the dimensions of the region.

```
#deixis-region-mark(  
  id: <celibate-region2>,  
  marker-position: (dx: 100% + 5pt, dy:  
  5pt))[#lorem(10)]
```

Default: auto

**stroke** auto or stroke or none

The border stroke of the region highlight.

Default: auto

**fill** auto or color or none

The background fill color of the region highlight.

Default: auto

**radius** auto or length or dictionary

The border radius of the region shape. Automatically adjusts when breaking across pages.

Default: auto

**padding** auto or length or str

The padding between the highlighted text and the border. Using "text" aligns to font metrics.

Default: `auto`

**region-shape** `auto` or `function`

Shape drawing function.

```
*Kepler's First Law of Planetary Motion*
states that every planet's orbit is an
ellipse:
#align(
  center,
  deixis-region-mark(padding: (x: 8pt, y:
8pt), region-shape: deixis-ellipse-
region)[$r = \frac{1 - e^2}{1 + e \cos(\theta)}$]
  cos(theta))$]
)
```

**Kepler's First Law of Planetary Motion**  
states that every planet's orbit is an ellipse:

$$r = \frac{1-e^2}{1+e \cos(\theta)}$$

Default: `auto`

**layer** `auto` or `str`

The rendering layer context. Choices: "flow" | "foreground" | "background".

- With "foreground" layer, the region is rendered in the foreground of the page, and may occlude the highlighted content.
- With "background" layer, the region will be rendered in the background of the page, and might be occluded by other contents.

```
// pins already placed in manual.typ
This example draws the #box(width: 0.6em,
height: 0.6em, stroke: yellow) yellow
region around this documentation chapter.
#deixis-region-mark(
  pins: ("region-mark-chapter-start",
"region-mark-chapter-end"),
  padding: (x: 5pt, top: -5pt, bottom:
5pt),
  stroke: yellow + 2pt,
  fill: yellow.transparentize(98%),
  radius: 10pt,
  layer: "background",
)
```

This example draws the  yellow region around this documentation chapter.

- With "flow" layer, the placement of the region is synchronuous.

```
#deixis-region-mark(
  pins: ("test-layer3", "test-layer4"),
  stroke: none,
  fill: purple.transparentize(25%),
  layer: "flow",
) // below the highlighted text

#quote(attribution: [Messmer the
Impaler], block: true)[*#deixis-
pin("test-layer1")Mongrel#deixis-
pin("test-layer2") #deixis-pin("test-
layer3")intruder#deixis-pin("test-
layer4").*]

#deixis-region-mark(
  pins: ("test-layer1", "test-layer2"),
  stroke: none,
  fill: red.transparentize(25%),
  layer: "flow",
) // above the highlighted text
```

Mongrel intruder.

— Messmer the Impaler

### 🔥 Tip

- Use "flow" for highlighting images, diagrams, ... Be aware that it will trigger a #parbreak.
- Use "background" for highlighting texts, equations, ... *if they are not in a non-transparent container.*
- Use "foreground" or "background" where the region can span across multiple pages.

Default: **auto**

**inline**    **bool**

If **true**, the region wraps the text in an inline box instead of a standard block.

Default: **false**

## 2.4. Notes

### 2.4.1. Inline Note

#### deixis-inline-note-body()

A standalone body content of an inline note.

```
The garden was filled with the
#deixis-inline-mark(id: <apricity>)
[apricity]
of a pale winter sun.

#deixis-inline-note-body(
  id: <apricity>,
  stroke: maroon,
  fill: maroon.transparentize(95%),
)[
  *Apricity:* The warmth of the sun in
  winter.
]
```

The garden was filled with the apricity<sup>1</sup> of a pale winter sun.

<sup>1</sup>**Apricity:** The warmth of the sun in winter.

#### Parameters

```
deixis-inline-note-body(
  body: content,
  id: none or str or label,
  target: auto or int or label or str,
  series: auto or str,
  label: none or label,
  marker-style: auto or function,
  backlink: auto or bool or str,
  inline-mode: auto or str,
  stroke: auto or stroke or none,
  fill: auto or color or none,
  radius: auto or length or dictionary,
  inline: bool,
  render-single: auto or function,
  container-func: auto or function,
  ..args: arguments
) -> content
```

**body**    content

The actual text or content of the note.

**id**    none or str or label

A unique identifier linking this body to its corresponding mark.

Default: none

**target**    auto or int or label or str

The specific block/minipage render context this note belongs to.

Default: auto

**series** `auto` or `str`

The counter series this note belongs to.

Default: `auto`

**label** `none` or `label`

An optional `label` to attach to the rendered body block.

Default: `none`

**marker-style** `auto` or `function`

A custom styling function (string) => content for the number/marker.

Default: `auto`

**backlink** `auto` or `bool` or `str`

Whether to generate a clickable backlink returning the reader to the mark. Choices: `bool` | `"always"` | `"none"` | `"never"` | `"multiple"`.

Default: `auto`

**inline-mode** `auto` or `str`

The visual decoration applied to the marked inline text. Choices: `"highlight"` | `"box"` | `"parentheses"` | `"underline"` | `"text-fill"` | `"none"`. Only used when `inline: true`.

Default: `auto`

**stroke** `auto` or `stroke` or `none`

The border stroke.

Default: `auto`

**fill** `auto` or `color` or `none`

The background fill color.

Default: `auto`

**radius** `auto` or `length` or `dictionary`

The border radius.

Default: `auto`

**inline** `bool`

If `true`, the body is rendered as if it is `deixis-inline-mark()`.

The city streets were filled with the psithurism `#deixis-inline-mark(id: <psithurism>)` of swaying trees. It was a comforting melody, `#deixis-inline-note-body(id: <psithurism>, inline: true)`[the sound of wind whispering through leaves].

The city streets were filled with the psithurism<sup>1</sup> of swaying trees. It was a comforting melody, [<sup>1</sup>the sound of wind whispering through leaves].

Default: `false`

**render-single** `auto` or `function`

A custom render function for the inner layout of the body.

Default: `auto`

**container-func** `auto` or `function`

A custom container wrapper for the body block.

Default: `auto`

**..args** `arguments`

Only accepts named arguments, which are forwarded to `container-func`.



## 2.4.2. Footnote

### deixis-footnote-body()

A standalone body content of a footnote.

The body of the footnote is rendered at the bottom of the page or minipage.

#### **i** Info

Since footnotes can interfere with the main layout, there is no easy way to implement a non-native footnote function. This *hacky* function internally calls `#std.footnote` if the note is page-level.

```
The evening sky was painted in a deep,
haunting
#deixis-inline-mark(id: <smalt>)[smalt]
that made the first stars appear like
diamonds on velvet.

#deixis-footnote-body(id: <smalt>)[A deep
blue pigment made from powdered glass
colored with cobalt oxide.]
```

The evening sky was painted in a deep, haunting  
**smalt**<sup>1</sup> that made the first stars appear like  
diamonds on velvet.

---

<sup>1</sup>A deep blue pigment made from powdered glass  
colored with cobalt oxide.

#### Parameters

```
deixis-footnote-body(
  body: content,
  id: none or str or label,
  target: auto or int or label or str,
  series: auto or str,
  label: none or label,
  marker-style: auto or function,
  backlink: auto or bool or str,
  stroke: auto or stroke or none,
  fill: auto or color or none,
  radius: auto or length or dictionary,
  render-single: auto or function,
  container-func: auto or function,
  ..args: arguments
) -> content
```

**body**    content

The actual text or content of the note.

**id**    none or str or label

A unique identifier linking this body to its corresponding mark.

Default: none

**target**    auto or int or label or str

The specific block/minipage render context this note belongs to.

Default: auto

**series** `auto` or `str`

The counter series this note belongs to.

Default: `auto`

**label** `none` or `label`

An optional `label` to attach to the rendered body block.

Default: `none`

**marker-style** `auto` or `function`

A custom styling function for the marker.

Default: `auto`

**backlink** `auto` or `bool` or `str`

Whether to generate a clickable backlink returning the reader to the mark. Choices: `bool` | `"always"` | `"none"` | `"never"` | `"multiple"`.

Default: `auto`

**stroke** `auto` or `stroke` or `none`

The border stroke.

Default: `auto`

**fill** `auto` or `color` or `none`

The background fill color.

Default: `auto`

**radius** `auto` or `length` or `dictionary`

The border radius.

Default: `auto`

**render-single** `auto` or `function`

A custom render function for the inner layout of the body.

Default: `auto`

**container-func** `auto` or `function`

A custom container wrapper for the body block.

Default: `auto`

**..args** arguments

Only accepts named arguments, which are forwarded to container-func.

### deixis-footnote()

A wrapper for generating footnotes.

This function creates a mark in the text and delegates the body to deixis-footnote-body().

The library's atmosphere was defined by a subtle  
`#deixis-footnote`[bibliosmia][The smell of  
old books.]  
that seemed to settle on the skin like a  
light dust.

The library's atmosphere was defined by a subtle  
**bibliosmia<sup>1</sup>** that seemed to settle on the skin  
like a light dust.

<sup>1</sup>The smell of old books.

### Parameters

```
deixis-footnote(  
    ..args: arguments ,  
    mark-type: auto str ,  
    id: none str label ,  
    numbering: auto str function ,  
    marker: auto content ,  
    target: int label str ,  
    series: str ,  
    label: none label ,  
    marker-style: auto function ,  
    marker-position: auto alignment dictionary ,  
    stroke: auto stroke none ,  
    fill: auto color none ,  
    radius: auto length dictionary ,  
    inline-mode: auto str ,  
    pins: array ,  
    padding: auto length str dictionary ,  
    region-shape: auto function ,  
    layer: auto str ,  
    inline: bool ,  
    backlink: auto bool str ,  
    render-single: auto function ,  
    container-func: auto function  
) -> content
```

**..args** arguments

Accepts up to two positional arguments: [mark][body]. All named arguments are forwarded to deixis-footnote-body().

**mark-type** auto or str

The type of mark to render in the text. Choices: "inline" | "region" | "phantom".

Default: auto

**id** `none` or `str` or `label`

A unique identifier linking an existing mark to the body internally. If `none`, a mark will be created.

Default: `none`

**numbering** `auto` or `str` or `function`

The numbering style for the marker.

Default: `auto`

**marker** `auto` or `content`

A hardcoded marker override.

Default: `auto`

**target** `int` or `label` or `str`

The specific block/minipage counter this note belongs to.

Default: `0`

**series** `str`

The counter series this note belongs to.

Default: `"default"`

**label** `none` or `label`

An optional `label` to attach to the rendered body block.

Default: `none`

**marker-style** `auto` or `function`

A custom styling function for the marker.

Default: `auto`

**marker-position** `auto` or `alignment` or `dictionary`

Mark marker placement.

See `deixis-inline-mark.marker-position` and `deixis-region-mark.marker-position`.

Default: `auto`

**stroke** `auto` or `stroke` or `none`

The border stroke.

Default: `auto`

**fill** `auto` or `color` or `none`

The background fill color.

Default: `auto`

**radius** `auto` or `length` or `dictionary`

The border radius.

Default: `auto`

**inline-mode** `auto` or `str`

The visual decoration applied to the marked inline text. Choices: `"highlight"` | `"box"` | `"parentheses"` | `"underline"` | `"text-fill"` | `"none"`.

See `deixis-inline-mark.inline-mode`.

Default: `auto`

**pins** `array`

Array of pin names for the region mark.

See `deixis-region-mark.pins`.

Default: `()`

**padding** `auto` or `length` or `str` or `dictionary`

Internal padding for the region mark.

See `deixis-region-mark.padding`.

Default: `auto`

**region-shape** `auto` or `function`

Shape drawing function for the region mark.

See `deixis-region-mark.region-shape`.

Default: `auto`

**layer** `auto` or `str`

Rendering layer for the region mark. Choices: `"flow"` | `"foreground"` | `"background"`.

See `deixis-region-mark.layer`.

Default: `auto`

**inline** `bool`

If `true`, force the region mark into an inline box.

See `deixis-region-mark.inline`.

Default: `false`

**backlink** `auto` or `bool` or `str`

Whether to generate a clickable backlink returning the reader to the mark. Choices: `bool` | `"always"` | `"none"` | `"never"` | `"multiple"`.

Default: `auto`

**render-single** `auto` or `function`

A custom render function for the inner layout of the body.

Default: `auto`

**container-func** `auto` or `function`

A custom container wrapper for the body block.

Default: `auto`

### 2.4.3. Endnote

#### deixis-endnote-body()

A standalone body content of an endnote.

The body of the endnote is completely invisible when declared. It is stored as metadata and will only be rendered when you explicitly call `#deixis-print-endnotes()`.

```
The abandoned cathedral was a monument to
#deixis-inline-mark(id: <sciamachy>,
series: "endnote")[sciamachy], its
crumbling walls silent witnesses to a
forgotten struggle.
Within its hallowed nave, a single #deixis-
inline-mark(id: <glabella>, series:
"endnote")[glabella] on a marble statue
caught the moonlight, the only smooth
surface left untouched by the decay of
time.
```

```
#deixis-endnote-body(id: <sciamachy>)
[*Sciamachy:* A battle against an imaginary
enemy; literally _shadow-fighting_.]
#deixis-endnote-body(id: <glabella>)
[*Glabella:* The smooth part of the
forehead between the eyebrows and just
above the nose.]
```

The abandoned cathedral was a monument to [sciamachy<sup>1</sup>], its crumbling walls silent witnesses to a forgotten struggle. Within its hallowed nave, a single [glabella<sup>2</sup>] on a marble statue caught the moonlight, the only smooth surface left untouched by the decay of time.

```
#deixis-print-endnotes()
```

- 1 **Sciamachy:** A battle against an imaginary enemy; literally *shadow-fighting*.
- 2 **Glabella:** The smooth part of the forehead between the eyebrows and just above the nose.

#### ! Memorize

It is noted that endnotes default to series: `"endnote"`. Hence, you should pass series: `"endnote"` to the decoupled mark or the body, or mind `deixis-print-endnotes.series` when you print them.

## Parameters

```
deixis-endnote-body(  
  body: content,  
  id: none or str or label,  
  target: auto or int or label or str,  
  series: auto or str,  
  label: none or label,  
  marker-style: auto or function,  
  backlink: auto or bool or str,  
  stroke: auto or stroke or none,  
  fill: auto or color or none,  
  radius: auto or length or dictionary,  
  render-single: auto or function,  
  container-func: auto or function,  
  ..args: arguments  
) -> content
```

**body**    `content`

The actual text or content of the note.

**id**    `none` or `str` or `label`

A unique identifier linking this body to its corresponding mark.

Default: `none`

**target**    `auto` or `int` or `label` or `str`

The specific block/minipage render context this note belongs to.

Default: `auto`

**series**    `auto` or `str`

The counter series this note belongs to.

Default: `auto`

**label**    `none` or `label`

An optional `label` to attach to the rendered body block.

Default: `none`

**marker-style**    `auto` or `function`

A custom styling function for the marker.

Default: `auto`

**backlink**    `auto` or `bool` or `str`

Whether to generate a clickable backlink returning the reader to the mark. Choices: `bool` | `"always"` | `"none"` | `"never"` | `"multiple"`.



Default: **auto**

**stroke** **auto** or **stroke** or **none**

The border stroke.

Default: **auto**

**fill** **auto** or **color** or **none**

The background fill color.

Default: **auto**

**radius** **auto** or **length** or **dictionary**

The border radius.

Default: **auto**

**render-single** **auto** or **function**

A custom render function for the inner layout of the body.

Default: **auto**

**container-func** **auto** or **function**

A custom container wrapper for the body block.

Default: **auto**

**..args** **arguments**

Only accepts named arguments, which are forwarded to `container-func`.

### **deixis-endnote()**

A wrapper for generating endnotes.

This function creates a mark in the text and delegates the body to `deixis-endnote-body()`.

```
The botanist spent years searching for the
idiosyncratic
#deixis-endnote[*Idiosyncratic:* Something
peculiar or individual to a specific
person, place, or thing.]
wildflower that only bloomed in the
valley's deepest shadows.
As the sun dipped below the horizon, the
forest floor began to glow with a soft
bioluminescence
#deixis-endnote[*Bioluminescence:* The
production and emission of light by a
living organism.]
that guided her back to camp.
```

The botanist spent years searching for the
idiosyncratic<sup>3</sup> wildflower that only bloomed in
the valley's deepest shadows. As the sun dipped
below the horizon, the forest floor began to glow
with a soft bioluminescence<sup>4</sup> that guided her
back to camp.

```
#deixis-print-endnotes()
```

- 3 **Idiosyncratic:** Something peculiar or individual to a specific person, place, or thing.
- 4 **Bioluminescence:** The production and emission of light by a living organism.

See `deixis-print-endnotes()`.

### Parameters

```
deixis-endnote(  
    ..args: arguments ,  
    mark-type: auto str ,  
    id: none str label ,  
    numbering: auto str function ,  
    marker: auto content ,  
    target: int label str ,  
    series: str ,  
    label: none label ,  
    marker-style: auto function ,  
    marker-position: auto alignment dictionary ,  
    stroke: auto stroke none ,  
    fill: auto color none ,  
    radius: auto length dictionary ,  
    inline-mode: auto str ,  
    pins: array ,  
    padding: auto length str dictionary ,  
    region-shape: auto function ,  
    layer: auto str ,  
    inline: bool ,  
    backlink: auto bool str ,  
    container-func: auto function  
) -> content
```

**..args** arguments

Accepts up to two positional arguments: [mark][body]. All named arguments are forwarded to `deixis-footnote-body()`.

**mark-type** auto or str

The type of mark to render in the text. Choices: "inline" | "region" | "phantom".

Default: auto

**id** none or str or label

A unique identifier linking an existing mark to the body internally. If none, a mark will be created.

Default: none

**numbering** auto or str or function

The numbering style for the marker.

Default: `auto`

**marker** `auto` or `content`

A hardcoded marker override.

Default: `auto`

**target** `int` or `label` or `str`

The specific block/minipage counter this note belongs to.

Default: `0`

**series** `str`

The counter series this note belongs to.

#### **i** Info

Note that endnotes belong to the `"endnote"` series by default, while all other note functions belong to the `"default"` series.

Default: `"endnote"`

**label** `none` or `label`

An optional `label` to attach to the rendered body block.

Default: `none`

**marker-style** `auto` or `function`

A custom styling function for the marker.

Default: `auto`

**marker-position** `auto` or `alignment` or `dictionary`

Mark marker placement.

See `deixis-inline-mark.marker-position` and `deixis-region-mark.marker-position`.

Default: `auto`

**stroke** `auto` or `stroke` or `none`

The border stroke.

Default: `auto`

**fill** `auto` or `color` or `none`

The background fill color.

Default: **auto**

**radius** **auto** or **length** or dictionary

The border radius.

Default: **auto**

**inline-mode** **auto** or **str**

The visual decoration applied to the marked inline text. Choices: **"highlight"** | **"box"** | **"parentheses"** | **"underline"** | **"text-fill"** | **"none"**.

See `deixis-inline-mark.inline-mode`.

Default: **auto**

**pins** array

Array of pin names for the region mark.

See `deixis-region-mark.pins`.

Default: **()**

**padding** **auto** or **length** or **str** or dictionary

Internal padding for the region mark.

See `deixis-region-mark.padding`.

Default: **auto**

**region-shape** **auto** or **function**

Shape drawing function for the region mark.

See `deixis-region-mark.region-shape`.

Default: **auto**

**layer** **auto** or **str**

Rendering layer for the region mark. Choices: **"flow"** | **"foreground"** | **"background"**.

See `deixis-region-mark.layer`.

Default: **auto**

**inline** **bool**

If **true**, force the region mark into an inline box.

See `deixis-region-mark.inline`.

Default: **false**

**backlink** `auto` or `bool` or `str`

Whether to generate a clickable backlink returning the reader to the mark. Choices: `bool` | `"always"` | `"none"` | `"never"` | `"multiple"`.

Default: `auto`

**container-func** `auto` or `function`

A custom container wrapper for the body block.

Default: `auto`

### `deixis-print-endnotes()`

Renders a collected list of endnote bodies at the current location in the document.

This function queries the deixis state engine for all endnotes matching the specified target and series, and prints them in a formatted list. It supports extensive spatial filtering, allowing you to print endnotes specifically for a single chapter or section.

#### **i** Info

Note bodies cannot be printed twice. `deixis-print-endnotes` keeps track of what was printed to avoid duplication.

```
#deixis-print-endnotes()
```

#### Parameters

```
deixis-print-endnotes(  
  target: int label str ,  
  series: auto str array ,  
  after: none location label str ,  
  before: none location label str ,  
  marker-style: auto function ,  
  body-style: auto function ,  
  backlink: auto bool str ,  
  separator: auto content none ,  
  clearance: auto length ,  
  gap: auto length ,  
  marker-gap: auto length ,  
  indent: auto length ,  
  render-single: auto function ,  
  render-group: auto function ,  
  container-func: auto function  
) -> content
```

**target** `int` or `label` or `str`

The specific block/minipage target to pull notes from.

Default: `0`

**series** `auto` or `str` or `array`

Filters to notes belonging to a specific series or array of series.

Default: `auto`

**after** `none` or `location` or `label` or `str`

Only prints notes that appear **after** this location, label, or `<deixis-pin>` name.

Default: `none`

**before** `none` or `location` or `label` or `str`

Only prints notes that appear **before** this location, label, or `<deixis-pin>` name.

Default: `none`

**marker-style** `auto` or `function`

Overrides the marker style for the printed list.

Default: `auto`

**body-style** `auto` or `function`

Overrides the text styling of the printed bodies.

Default: `auto`

**backlink** `auto` or `bool` or `str`

Overrides the backlink generation setting for these notes. Choices: `bool` | `"always"` | `"none"` | `"never"` | `"multiple"`.

Default: `auto`

**separator** `auto` or `content` or `none`

An optional separator to draw between the list of notes (e.g., a line).

Default: `auto`

**clearance** `auto` or `length`

The spacing before the list of endnotes begins.

Default: `auto`

**gap** `auto` or `length`

The vertical spacing between individual endnotes.

Default: `auto`

**marker-gap** `auto` or `length`

The horizontal spacing between the endnote number and its text body.

Default: `auto`

**indent** `auto` or `length`

Overrides the indentation applied to the note body.

Default: `auto`

**render-single** `auto` or `function`

A custom rendering function (note-data) => content to override the local render-single.

Default: `auto`

**render-group** `auto` or `function`

A custom rendering function (notes-array) => content to override the local render-group.

Default: `auto`

**container-func** `auto` or `function`

A custom container wrapper for the entire printed block.

Default: `auto`

## 2.4.4. Margin Note

### deixis-margin-note-body()

A standalone body content of a margin note.

```
The old manor was a labyrinth of
#deixis-inline-mark(id: <crepuscular>)
[crepuscular]
corridors, where the shadows seemed to
stretch and breathe with the setting sun.

#deixis-margin-note-body(id: <crepuscular>)
[Relating to or resembling twilight.]
```

The old manor was a labyrinth of crepuscular<sup>14</sup> corridors, where the shadows seemed to stretch and breathe with the setting sun.

<sup>14</sup>Relating to or resembling twilight.

### Parameters

```
deixis-margin-note-body(
  body: content,
  id: none | str | label,
  target: auto | int | label | str,
  series: auto | str,
  label: none | label,
  marker-style: auto | function,
  backlink: auto | bool | str,
  stroke: auto | stroke | none,
  fill: auto | color | none,
  radius: auto | length | dictionary,
  link: auto | str,
  link-waypoints: none | auto | array,
  link-ports: auto | array | dictionary,
  link-marks: auto | str,
  width: auto | length | relative,
  dx: auto | length,
  dy: length,
  side: auto | alignment,
  side-strategy: auto | str,
  mark-align: auto | alignment | dictionary,
  mark-align-strictness: auto | str,
  spillover: auto | bool,
  anchor-pin: none | str,
  render-single: auto | function,
  container-func: auto | function,
  ..args: arguments
) -> content
```

**body**    content

The actual text or content of the note.

**id**    none or str or label

A unique identifier linking this body to its corresponding mark.

Default: none



**target** `auto` or `int` or `label` or `str`

The specific block/minipage render context this note belongs to.

Default: `auto`

**series** `auto` or `str`

The counter series this note belongs to.

Default: `auto`

**label** `none` or `label`

An optional `label` to attach to the rendered body block.

Default: `none`

**marker-style** `auto` or `function`

A custom styling function for the marker.

Default: `auto`

**backlink** `auto` or `bool` or `str`

Whether to generate a clickable backlink returning the reader to the mark. Choices: `bool` | `"always"` | `"none"` | `"never"` | `"multiple"`.

Default: `auto`

**stroke** `auto` or `stroke` or `none`

The border stroke.

Default: `auto`

**fill** `auto` or `color` or `none`

The background fill color.

Default: `auto`

**radius** `auto` or `length` or `dictionary`

The border radius.

Default: `auto`

**link** `auto` or `str`

The type of connector line. Choices: `"none"` | `"straight-line"` | `"right-angle"` | `"chamfer"` | `"curve"` | `"ucr"` | `"ccr"`.

```
#deixis-margin-note(link: "right-angle")  
[A linked note.]
```

15

<sup>15</sup> A linked note.

Default: **auto**

**link-waypoints** **none** or **auto** or **array**

An array of intermediate coordinate offsets to route the connector line through.

Default: **auto**

**link-ports** **auto** or **array** or **dictionary**

Defines the exact exit/entry ports on the mark and body for the connector line.

Default: **auto**

**link-marks** **auto** or **str**

The arrowhead style for the connector line. Choices: **"none"** | **"mark"** | **"body"** | **"both"**.

Default: **auto**

**width** **auto** or **length** or **relative**

The width of the margin note body. Defaults to **90%** of the margin width.

```
#lorem(3)  
#deixis-margin-note(width: 50%) [A narrow  
50% note.]
```

Lorem ipsum dolor.<sup>16</sup>

<sup>16</sup> A narrow 50% note.

Default: **90%**

**dx** **auto** or **length**

Horizontal offset adjustment for the body.

```
#deixis-margin-note(dx: -30pt) [A  
horizontally shifted note.]
```

17

<sup>17</sup> A horizontally shifted note.

Default: **auto**

**dy** **length**

Vertical offset adjustment for the body.

### **i** Info

dy is applied pre-layout, meaning that the final coordinates of the note can change after the rubber band pass.

```
#deixis-margin-note(dy: -50pt)[A  
vertically shifted note.]
```

18

<sup>18</sup>A vertically shifted note.

Default: 0pt

**side** **auto** or **alignment**

Which margin to place the note in (left or right). If **auto**, uses the side nearest the mark.

```
#deixis-margin-note(side: left)[This is a  
left side note.]  
#deixis-margin-note(side: right)[This is  
a right side note.]
```

1920

<sup>19</sup>This is a left side note.

<sup>20</sup>This is a right side note.

Default: **auto**

**side-strategy** **auto** or **str**

Strategy to select side for margin note. Choices: "nearest" | "strict".

- With side-strategy: "nearest", notes will prefer the side that minimizes the vertical distance from the mark to the body.

```
#deixis-margin-note(side-strategy:  
"nearest")[Note A body.]  
#deixis-margin-note(side-strategy:  
"nearest")[Note B body.]
```

2122

<sup>22</sup>Note B body.

<sup>21</sup>Note A body.

- With side-strategy: "strict", notes will prefer the more visually favorable side until it is overflow.

```
#deixis-margin-note(side-strategy:  
"strict")[Note A body.]  
#deixis-margin-note(side-strategy:  
"strict")[Note B body.]
```

2324

<sup>23</sup>Note A body.

<sup>24</sup>Note B body.

Default: **auto**

**mark-align** **auto** or **alignment** or **dictionary**

How the note aligns vertically relative to its mark.

- It can be an **alignment** or an equivalent **str**.

```
#lorem(8)  
#deixis-margin-note(  
  mark-align: bottom,  
  link: "right-angle",  
  stroke: red,  
  container-func: deixis-rect-container)  
[Top-aligned note body]
```

Lorem ipsum dolor sit amet, consectetur  
adipiscing elit<sup>25</sup>

<sup>25</sup>Top-aligned  
note body

- Or a component-scope **dictionary**, only useful with mark-type: "region".

<sup>28</sup>This is another extremely long note. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliquam quaerat voluptatem. Ut enim aequae doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facere et urbane Stoicos irridere, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum.

```
#deixis-margin-note(  
  mark-type: "region",  
  padding: "text",  
  mark-align: (mark: horizon, body:  
horizon),  
  link: "right-angle",  
  stroke: red,  
  container-func: deixis-rect-container)  
[#lorem(8)][Horizon-aligned note body.]
```

Default: **auto**

**mark-align-strictness**   **auto** or **str**

How strictly the engine attempts to keep the note exactly aligned with its mark. Choices: **"strict"** | **"loose"** | **"none"**.

Default: **auto**

**spillover**   **auto** or **bool**

Allow this note to spill over page boundary.

```
#deixis-margin-note(spillover: false)  
[This is a very long note. #lorem(120)]  
#deixis-margin-note(spillover: false)  
[This is another extremely long note.  
#lorem(120)]  
#deixis-margin-note(spillover: true)[This  
is a spilled note.]
```

Default: **auto**

**anchor-pin**   **none** or **str**

A specific #deixis-pin name to anchor the body to, instead of the default layout logic.

```
#deixis-margin-note(  
  anchor-pin: "pin0") [A note anchored to  
a pin.]  
#lorem(20)  
#deixis-pin("pin0")
```

Default: **none**

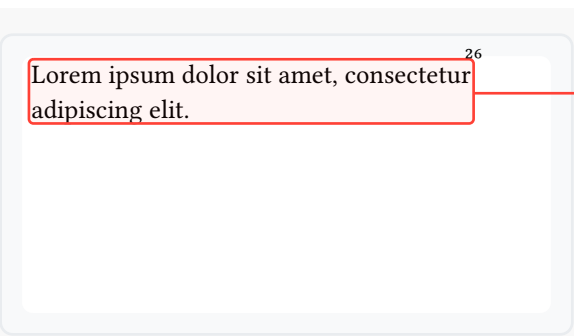
**render-single**   **auto** or **function**

A custom render function for the inner layout of the body.

Default: **auto**

**container-func**   **auto** or **function**

A custom container wrapper for the body block.



<sup>26</sup>Horizon-aligned note body.

<sup>27</sup>This is a very long note. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliquam quaerat voluptatem. Ut enim aequae doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facere et urbane Stoicos irridere, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum.

<sup>30</sup>A note anchored to a pin.

Default: `auto`

<sup>29</sup>This is a spilled note.

`..args` arguments

Only accepts named arguments, which are forwarded to `container-func`.

### `deixis-margin-note()`

A wrapper for generating margin notes.

This function creates a mark in the text and delegates the body to `deixis-margin-note-body()`.

The traveler felt a sudden sense of  
`#deixis-margin-note[fernweh]`[A deep, aching  
longing for far-off places; literally  
`_farsickness_`.]  
as he watched the ships disappear beyond  
the horizon, bound for lands he had never  
seen.

The traveler felt a sudden sense of `fernweh`<sup>31</sup> as  
he watched the ships disappear beyond the  
horizon, bound for lands he had never seen.

<sup>31</sup>A deep, aching  
longing for far-  
off places; liter-  
ally *farsickness*.

## Parameters

```
deixis-margin-note(  
  ..args: arguments ,  
  mark-type: auto str ,  
  id: none str label ,  
  numbering: auto str function ,  
  marker: auto content ,  
  target: int label str ,  
  series: str ,  
  label: none label ,  
  marker-style: auto function ,  
  marker-position: auto alignment dictionary ,  
  stroke: auto stroke none ,  
  fill: auto color none ,  
  radius: auto length dictionary ,  
  inline-mode: auto str ,  
  pins: array ,  
  padding: auto length str dictionary ,  
  region-shape: auto function ,  
  layer: auto str ,  
  inline: bool ,  
  backlink: auto bool str ,  
  link: auto str ,  
  link-waypoints: none auto array ,  
  link-ports: auto array dictionary ,  
  link-marks: auto str ,  
  width: auto length relative ,  
  dx: auto length ,  
  dy: length ,  
  side: auto alignment ,  
  side-strategy: auto str ,  
  mark-align: auto alignment dictionary ,  
  mark-align-strictness: auto str ,  
  spillover: auto bool ,  
  anchor-pin: none str ,  
  render-single: auto function ,  
  container-func: auto function  
) -> content
```

**..args** arguments

Accepts up to two positional arguments: [mark][body]. All named arguments are forwarded to `deixis-margin-note-body()`.

**mark-type** auto or str

The type of mark to render in the text. Choices: "inline" | "region" | "phantom".

Default: auto

**id** none or str or label

A unique identifier linking an existing mark to the body internally. If none, a mark will be created.

Default: `none`

**numbering** `auto` or `str` or `function`

The numbering style for the marker.

Default: `auto`

**marker** `auto` or `content`

A hardcoded marker override.

Default: `auto`

**target** `int` or `label` or `str`

The specific block/minipage counter this note belongs to.

Default: `0`

**series** `str`

The counter series this note belongs to.

Default: `"default"`

**label** `none` or `label`

An optional `label` to attach to the rendered body block.

Default: `none`

**marker-style** `auto` or `function`

A custom styling function for the marker.

Default: `auto`

**marker-position** `auto` or `alignment` or `dictionary`

Mark marker placement.

See `deixis-inline-mark.marker-position` and `deixis-region-mark.marker-position`.

Default: `auto`

**stroke** `auto` or `stroke` or `none`

The border stroke.

Default: `auto`

**fill** `auto` or `color` or `none`

The background fill color.

Default: **auto**

**radius** **auto** or **length** or dictionary

The border radius.

Default: **auto**

**inline-mode** **auto** or **str**

The visual decoration applied to the marked inline text. Choices: **"highlight"** | **"box"** | **"parentheses"** | **"underline"** | **"text-fill"** | **"none"**.

See `deixis-inline-mark.inline-mode`.

Default: **auto**

**pins** array

Array of pin names for the region mark.

See `deixis-region-mark.pins`.

Default: **()**

**padding** **auto** or **length** or **str** or dictionary

Internal padding for the region mark.

See `deixis-region-mark.padding`.

Default: **auto**

**region-shape** **auto** or **function**

Shape drawing function for the region mark.

See `deixis-region-mark.region-shape`.

Default: **auto**

**layer** **auto** or **str**

Rendering layer for the region mark. Choices: **"flow"** | **"foreground"** | **"background"**.

See `deixis-region-mark.layer`.

Default: **auto**

**inline** **bool**

If **true**, force the region mark into an inline box.

See `deixis-region-mark.inline`.

Default: **false**



**backlink** `auto` or `bool` or `str`

Whether to generate a clickable backlink returning the reader to the mark. Choices: `bool` | `"always"` | `"none"` | `"never"` | `"multiple"`.

Default: `auto`

**link** `auto` or `str`

The type of connector line. Choices: `"none"` | `"straight-line"` | `"right-angle"` | `"chamfer"` | `"curve"` | `"ucr"` | `"ccr"`.

See `deixis-margin-note-body.link`.

Default: `auto`

**link-waypoints** `none` or `auto` or `array`

An array of intermediate coordinate offsets to route the connector line through.

Default: `auto`

**link-ports** `auto` or `array` or `dictionary`

Defines the exact exit/entry ports on the mark and body for the connector line.

Default: `auto`

**link-marks** `auto` or `str`

The arrowhead style for the connector line. Choices: `"none"` | `"mark"` | `"body"` | `"both"`.

Default: `auto`

**width** `auto` or `length` or `relative`

The width of the margin note body.

See `deixis-margin-note-body.width`.

Default: `90%`

**dx** `auto` or `length`

Horizontal offset adjustment for the body.

See `deixis-margin-note-body.dx`.

Default: `auto`

**dy** `length`

Vertical offset adjustment for the body.

See `deixis-margin-note-body.dy`.

Default: `0pt`

**side** `auto` or `alignment`

Which margin to place the note in (left or right).

See `deixis-margin-note-body.side`.

Default: `auto`

**side-strategy** `auto` or `str`

Strategy to select side for margin note. Choices: `"nearest"` | `"strict"`.

See `deixis-margin-note-body.side-strategy`.

Default: `auto`

**mark-align** `auto` or `alignment` or `dictionary`

How the note aligns vertically relative to its mark.

See `deixis-margin-note-body.mark-align`.

Default: `auto`

**mark-align-strictness** `auto` or `str`

How strictly the engine attempts to keep the note exactly aligned with its mark. Choices: `"strict"` | `"loose"` | `"none"`.

See `deixis-margin-note-body.mark-align-strictness`.

Default: `auto`

**spillover** `auto` or `bool`

Allow this note to spill over page boundary.

See `deixis-margin-note-body.spillover`.

Default: `auto`

**anchor-pin** `none` or `str`

A specific `#deixis-pin` name to anchor the body to, instead of the default layout logic.

See `deixis-margin-note-body.anchor-pin`.

Default: `none`

**render-single** `auto` or `function`

A custom render function for the inner layout of the body.

Default: `auto`

**container-func** `auto` or `function`

A custom container wrapper for the body block.

Default: **auto**

### 2.4.5. Inset Note

#### deixis-inset-note-body()

A standalone body content of an inset note.

The body of inset notes are placed at precise geometric coordinates relative an anchor (its mark or a specific pin). They are perfect for callouts.

```
The artist spent months wandering the dense
pine forests, completely captivated by the
local #deixis-inline-mark(id:
<apricharity>)[apricharity].
```

```
#deixis-inset-note-body(
  id: <apricharity>,
  dx: 2.5em,
  dy: 2em,
)[
  warmth of the sun in winter.
]
```

The artist spent months wandering the dense  
pine forests, completely captivated by the local [ apricharity<sup>1</sup>.

<sup>1</sup>warmth of the sun in winter.

#### Parameters

```
deixis-inset-note-body(
  body: content,
  id: none str label,
  target: auto int label str,
  series: auto str,
  label: none label,
  marker-style: auto function,
  backlink: auto bool str,
  stroke: auto stroke none,
  fill: auto color none,
  radius: auto length dictionary,
  link: auto str,
  link-waypoints: none array,
  link-ports: auto array dictionary,
  link-marks: auto str,
  width: auto length relative,
  placement: none function,
  dx: none length,
  dy: none length,
  anchor: dictionary,
  anchor-pin: none str,
  layer: auto str,
  render-single: auto function,
  container-func: auto function,
  ..args: arguments
) -> content
```

**body** content

The actual text or content of the note.

**id** none or str or label

A unique identifier linking this body to its corresponding mark.

Default: `none`

**target** `auto` or `int` or `label` or `str`

The specific block/minipage render context this note belongs to.

Default: `auto`

**series** `auto` or `str`

The counter series this note belongs to.

Default: `auto`

**label** `none` or `label`

An optional `label` to attach to the rendered body block.

Default: `none`

**marker-style** `auto` or `function`

A custom styling function for the marker.

Default: `auto`

**backlink** `auto` or `bool` or `str`

Whether to generate a clickable backlink returning the reader to the mark. Choices: `bool` | `"always"` | `"none"` | `"never"` | `"multiple"`.

Default: `auto`

**stroke** `auto` or `stroke` or `none`

The border stroke.

Default: `auto`

**fill** `auto` or `color` or `none`

The background fill color.

Default: `auto`

**radius** `auto` or `length` or `dictionary`

The border radius.

Default: `auto`

**link** `auto` or `str`

The type of connector line. Choices: "none" | "straight-line" | "right-angle" | "chamfer" | "curve" | "ucr" | "ccr".

Default: **auto**

**link-waypoints** **none** or **array**

An array of intermediate coordinate offsets to route the connector line through.

Default: **none**

**link-ports** **auto** or **array** or **dictionary**

Defines the exact exit/entry ports for the connector line.

Default: **auto**

**link-marks** **auto** or **str**

The arrowhead style for the connector line. Choices: "none" | "mark" | "body" | "both".

Default: **auto**

**width** **auto** or **length** or **relative**

The width of the inset note body.

Default: **auto**

**placement** **none** or **function**

A custom floating placement function (e.g., Typst's native place).

```
#let place-func(it) = {  
  std.place(top + right, it)  
}  
#deixis-inset-note(placement: place-func)  
[Mark][Inset note placed by `placement`.]
```

Mark<sup>1</sup>

<sup>1</sup>Inset note placed by placement.

Default: **none**

**dx** **none** or **length**

Absolute horizontal offset for placing the note.

```
#deixis-inset-note(dx: 10pt, dy: 10pt)  
[Mark][Inset note placed with `dx`,  
`dy`.]
```

Mark<sup>1</sup>

<sup>1</sup>Inset note placed with dx, dy.

Default: **none**

**dy** **none** or **length**

Absolute vertical offset for placing the note.

See `deixis-inset-note.dx`.

Default: `none`

**anchor** `dictionary`

A dictionary defining how the note aligns to its anchor point (e.g., (mark: horizon + right, body: horizon + left)).

```
#deixis-inset-note(  
  anchor: (mark: bottom + right, body:  
    top + left),  
  dx: 0pt, dy: 0pt)[Mark][Inset note  
  anchored to the mark.]
```

Mark<sup>1</sup>

<sup>1</sup>Inset note anchored to the mark.

Default: (mark: horizon + right, body: horizon + left)

**anchor-pin** `none` or `str`

A specific `#deixis-pin` name to anchor the inset note to.

```
#deixis-inset-note(  
  anchor-pin: "anchor-pin",  
  anchor: (mark: horizon, body: horizon +  
    left),  
  layer: "flow",  
  dx: 0pt, dy: 0pt)[Mark][Inset note  
  anchored to a pin.]  
  
#lorem(3)#deixis-pin("anchor-pin")
```

Mark<sup>1</sup>

Lorem ipsum dolor. <sup>1</sup>Inset note anchored to a pin.

See `deixis-pin()`.

Default: `none`

**layer** `auto` or `str`

The rendering layer context. Choices: `"flow"` | `"foreground"` | `"background"`.

#### ⚠ Warning

Similar to `deixis-region-mark.layer`, the `"flow"` layer will trigger a `#parbreak` and disrupt your paragraph. However, it is still recommended to use `layer: "flow"` whenever possible for stability.

Default: `"flow"`

**render-single** `auto` or `function`

A custom render function for the inner layout of the body.

Default: `auto`

**container-func** `auto` or `function`

A custom container wrapper for the body block.

Default: `auto`

**..args** `arguments`

Only accepts named arguments, which are forwarded to `container-func`.

### `deixis-inset-note()`

A wrapper for generating inset notes.

This function creates a mark in the text and delegates the body to `deixis-inset-note-body()`.

```
The city council has voted to pave the
trail with
#deixis-inset-note(dx: 10pt, dy: 30pt,
width: 100pt)[
  permeable pavement
][
  A porous surface that lets rainwater
filter directly into the ground.
].
```

The city council has voted to pave the trail with [ permeable pavement<sup>1</sup>].

<sup>1</sup>A porous surface that lets rainwater filter directly into the ground.



## Parameters

```
deixis-inset-note(  
  ..args: arguments ,  
  mark-type: auto str ,  
  id: none str label ,  
  numbering: auto str function ,  
  marker: auto content ,  
  target: int label str ,  
  series: str ,  
  label: none label ,  
  marker-style: auto function ,  
  stroke: auto stroke none ,  
  fill: auto color none ,  
  radius: auto length dictionary ,  
  marker-position: auto alignment dictionary ,  
  inline-mode: auto str ,  
  pins: array ,  
  padding: auto length str dictionary ,  
  region-shape: auto function ,  
  inline: bool ,  
  backlink: auto bool str ,  
  link: auto str ,  
  link-waypoints: none array ,  
  link-ports: auto array dictionary ,  
  link-marks: auto str ,  
  width: auto length relative ,  
  placement: none function ,  
  dx: none length ,  
  dy: none length ,  
  anchor: dictionary ,  
  anchor-pin: none str ,  
  layer: auto str dictionary ,  
  render-single: auto function ,  
  container-func: auto function  
) -> content
```

**..args** arguments

Accepts up to two positional arguments: [mark][body]. All named arguments are forwarded to `deixis-inset-note-body()`.

**mark-type** auto or str

The type of mark to render in the text. Choices: "inline" | "region" | "phantom".

Default: auto

**id** none or str or label

A unique identifier linking an existing mark to the body internally. If none, a mark will be created.

Default: none

**numbering** `auto` or `str` or `function`

The numbering style for the marker.

Default: `auto`

**marker** `auto` or `content`

A hardcoded marker override.

Default: `auto`

**target** `int` or `label` or `str`

The specific block/minipage counter this note belongs to.

Default: `0`

**series** `str`

The counter series this note belongs to.

Default: `"default"`

**label** `none` or `label`

An optional `label` to attach to the rendered body block.

Default: `none`

**marker-style** `auto` or `function`

A custom styling function for the marker.

Default: `auto`

**stroke** `auto` or `stroke` or `none`

The border stroke.

Default: `auto`

**fill** `auto` or `color` or `none`

The background fill color.

Default: `auto`

**radius** `auto` or `length` or `dictionary`

The border radius.

Default: `auto`

**marker-position** `auto` or `alignment` or `dictionary`

Mark marker placement.

See `deixis-inline-mark.marker-position` and `deixis-region-mark.marker-position`.

Default: `auto`

**inline-mode** `auto` or `str`

The visual decoration applied to the marked inline text. Choices: `"highlight"` | `"box"` | `"parentheses"` | `"underline"` | `"text-fill"` | `"none"`.

See `deixis-inline-mark.inline-mode`.

Default: `auto`

**pins** `array`

Array of pin names for the region mark.

See `deixis-region-mark.pins`.

Default: `()`

**padding** `auto` or `length` or `str` or `dictionary`

Internal padding for the region mark.

See `deixis-region-mark.padding`.

Default: `auto`

**region-shape** `auto` or `function`

Shape drawing function for the region mark.

See `deixis-region-mark.region-shape`.

Default: `auto`

**inline** `bool`

If `true`, force the region mark into an inline box.

See `deixis-region-mark.inline`.

Default: `false`

**backlink** `auto` or `bool` or `str`

Whether to generate a clickable backlink returning the reader to the mark. Choices: `bool` | `"always"` | `"none"` | `"never"` | `"multiple"`.

Default: `auto`

**link** `auto` or `str`

The type of connector line. Choices: `"none"` | `"straight-line"` | `"right-angle"` | `"chamfer"` | `"curve"` | `"ucr"` | `"ccr"`.

Default: `auto`

**link-waypoints** `none` or `array`

An array of intermediate coordinate offsets to route the connector line through.

Default: `none`

**link-ports** `auto` or `array` or `dictionary`

Defines the exact exit/entry ports for the connector line.

Default: `auto`

**link-marks** `auto` or `str`

The arrowhead style for the connector line. Choices: `"none"` | `"mark"` | `"body"` | `"both"`.

Default: `auto`

**width** `auto` or `length` or `relative`

The width of the inset note body.

Default: `auto`

**placement** `none` or `function`

A custom floating placement function (e.g., Typst's native `#std.place`).

See `deixis-inset-note-body.placement`.

Default: `none`

**dx** `none` or `length`

Absolute horizontal offset for placing the note.

See `deixis-inset-note-body.dx`.

Default: `none`

**dy** `none` or `length`

Absolute vertical offset for placing the note.

See `deixis-inset-note-body.dy`.

Default: `none`

**anchor** `dictionary`

A dictionary defining how the note aligns to its anchor point (e.g., (mark: horizon + right, body: horizon + left)).

See `deixis-inset-note-body.anchor`.

Default: (mark: horizon + right, body: horizon + left)

**anchor-pin** `none` or `str`

A specific `#deixis-pin` name to anchor the inset note to.

See `deixis-inset-note-body.anchor-pin`.

Default: `none`

**layer** `auto` or `str` or `dictionary`

Rendering layer for the region mark and/or the body. Choices: `"flow"` | `"foreground"` | `"background"`.

See `deixis-region-mark.layer` and `deixis-inset-note-body.layer`.

#### Tip

Since both `deixis-region-mark()` and `deixis-inset-note-body()` shares this parameter, you can pass `layer: (mark: "flow", body: "foreground")` to assign different layer to each component.

Default: `auto`

**render-single** `auto` or `function`

A custom render function for the inner layout of the body.

Default: `auto`

**container-func** `auto` or `function`

A custom container wrapper for the body block.

Default: `auto`

## 2.5. Renderers

### 2.5.1. Note Renderer

#### `deixis-native-render-single()`

A rendering function that formats the note body closely matching Typst's native footnote style.

##### Parameters

```
deixis-native-render-single(  
  data,  
  inner-width,  
  ..args  
) -> content
```

#### `deixis-default-render-single()`

The default rendering function, formatting the note body with the marker aligned to the side.

##### Parameters

```
deixis-default-render-single(  
  data,  
  inner-width,  
  max-marker-width,  
  max-marker-gap,  
  ..args  
) -> content
```

### 2.5.2. Group Renderer

#### `deixis-native-render-group()`

Render a group of notes by vertical stacking, similar to Typst's native footnote stacking.

##### Parameters

```
deixis-native-render-group(  
  notes-data,  
  gap,  
  ..args  
) -> content
```

#### `deixis-default-render-group()`

Render a group of notes by vertical stacking while aligning body markers. Should be used in combination with `deixis-default-render-single()`.

##### Parameters

```
deixis-default-render-group(  
  notes-data,  
  gap,  
  ..args  
) -> content
```

#### `deixis-grid-render-group()`

Render a group of notes by putting them into a grid layout.

#### **i** Info

Unlike other group renderers, this function does not forward call to individual `render-single` of each note.

### Parameters

```
deixis-grid-render-group(  
  notes-data,  
  gap,  
  ..args  
) -> content
```

### 2.5.3. Helpers

These helper functions can be useful for implementing a custom render function.

#### **deixis-generate-backlinks()**

Returns the backlink buttons linking to the mark and anywhere the body is referenced.

### Parameters

```
deixis-generate-backlinks(data) -> content
```

#### **deixis-generate-body-meta()**

Returns the invisible metadata allowing other components to link to the note body, usually put before the body marker.

### Parameters

```
deixis-generate-body-meta(data) -> content
```

#### **deixis-generate-body-marker()**

Returns the visual marker for the note body, attaching metadata from `deixis-generate-body-meta()`.

### Parameters

```
deixis-generate-body-marker(data) -> content
```

## 2.6. Minipage

### deixis-block()

Creates an isolated, self-contained layout environment for notes.

A minipage acts as a sub-document. Any notes declared inside it (like margin notes or footnotes) are strictly bound to its physical boundaries, and its note counters operate independently of the main document flow. This is essential for creating complex layouts like infoboxes, sidebars, or custom chapter summaries without breaking global note numbering.

```
#deixis-footnote[This is a page footnote.]
#deixis-block(inset: 5pt, fill:
gray.lighten(90%)) [
  Inside a `#deixis-block`, notes have its
  own counter system and layout behaviors.
  #deixis-footnote[This is a block-level
  footnote.]
]
```

32

Inside a #deixis-block, notes have its own counter system and layout behaviors.<sup>1</sup>

<sup>1</sup>This is a block-level footnote.

### Experiment

You can nest #deixis-block, infinitely, but it will be hyper fragile.

```
#deixis-block(inset: 5pt, fill:
red.lighten(90%)) [
  #deixis-footnote[Level 1.]
  #deixis-block(inset: 5pt, fill:
orange.lighten(90%)) [
    #deixis-footnote[Level 2.]
    #deixis-block(inset: 5pt, fill:
yellow.lighten(90%)) [
      #deixis-footnote[Level 3.]
      #deixis-block(inset: 5pt, fill:
green.lighten(90%)) [
        You are wandering into the
        unknown#deixis-footnote[Level 4.].
      ]
    ]
  ]
]
```

1

1

1

You are wandering into the  
unknown<sup>1</sup>.

<sup>1</sup>Level 4.

<sup>1</sup>Level 3.

<sup>1</sup>Level 2.

<sup>1</sup>Level 1.

<sup>32</sup>This is a page footnote.



## Parameters

```
deixis-block(  
  id: auto str,  
  thread: none str,  
  sync-counters-with: auto int label str,  
  width: auto length relative,  
  height: auto length relative,  
  breakable: auto bool,  
  fill: auto color none,  
  stroke: auto stroke none,  
  radius: auto length dictionary,  
  inset: auto length dictionary,  
  outset: auto length dictionary,  
  spacing: auto length,  
  above: auto length,  
  below: auto length,  
  clip: auto bool,  
  sticky: auto bool,  
  ..args: arguments  
) -> content
```

**id** auto or str

A unique identifier for this layout context. If **auto**, the engine generates one automatically.

Default: **auto**

**thread** none or str

Allows multiple separate minipages to share the same continuous note counter. Pass a matching string to link them into the same “thread”.

```
#deixis-block(thread: "green", inset:  
5pt, fill: green.lighten(90%))[  
  #deixis-footnote[This block belongs to  
the `green` thread.]  
]  
  
#deixis-block(inset: 5pt, fill:  
red.lighten(90%))[  
  #deixis-footnote[This block does not,  
hence, its notes count from 1.]  
]  
  
#deixis-block(thread: "green", inset:  
5pt, fill: green.lighten(90%))[  
  #deixis-footnote[This block also  
belongs to the `green` thread.]  
]
```

1  
\_\_\_\_\_

<sup>1</sup>This block belongs to the "green" thread.

1  
\_\_\_\_\_

<sup>1</sup>This block does not, hence, its notes count from 1.

2  
\_\_\_\_\_

<sup>2</sup>This block also belongs to the "green" thread.

### Tip

thread cannot be used to sync counters with the "page". For that purpose, use `deixis-block.sync-counters-with`.

### Experiment

thread and sync-counters-with can be used together, at your own risk.

Default: **none**

### **sync-counters-with** auto or int or label or str

Synchronize internal counters with another `#deixis-block` or the global document counters rather than starting from 1.

- Accepts int for going up in the stack level, and "page" for the document global.

```
#deixis-block(sync-counters-with: "page",
inset: 5pt, fill: red.lighten(90%))[
  #deixis-footnote[This block shares its
counters with the document.]
]

#deixis-block(sync-counters-with: -1,
inset: 5pt, fill: orange.lighten(90%))[
  #deixis-footnote[Same for this block.]
]
```

33

<sup>33</sup>This block shares its counters with the document.

34

<sup>34</sup>Same for this block.

- Also accepts label or str for targeting another `#deixis-block`.

```
#deixis-block(id: <blue-block>, inset:
5pt, fill: blue.lighten(90%))[
  #deixis-footnote[This block is given an
`id: <blue-block>`.]
]

#deixis-block(sync-counters-with: <blue-
block>, inset: 5pt, fill:
teal.lighten(90%))[
  #deixis-footnote[This block shares its
counters with `<blue-block>`.]
]
```

1

<sup>1</sup>This block is given an id: <blue-block>.

2

<sup>2</sup>This block shares its counters with <blue-block>.

### Tip

To sync counters of multiple blocks, it is more convenient to use `deixis-block.thread`.

Default: **auto**

### **width** auto or length or relative

See `std.block.width`.

Default: `auto`

**height** `auto` or `length` or `relative`

See `std.block.height`.

Default: `auto`

**breakable** `auto` or `bool`

See `std.block.breakable`.

Default: `auto`

**fill** `auto` or `color` or `none`

See `std.block.fill`.

Default: `auto`

**stroke** `auto` or `stroke` or `none`

See `std.block.stroke`.

Default: `auto`

**radius** `auto` or `length` or `dictionary`

See `std.block.radius`.

Default: `auto`

**inset** `auto` or `length` or `dictionary`

See `std.block.inset`.

#### **i** Info

The left and right inset of a `#deixis-block` is where block-level `deixis-margin-note-body()` are rendered.

Default: `auto`

**outset** `auto` or `length` or `dictionary`

See `std.block.outset`.

Default: `auto`

**spacing** `auto` or `length`

See `std.block.spacing`.

Default: `auto`

**above** `auto` or `length`

See `std.block.above`.

Default: `auto`

**below** `auto` or `length`

See `std.block.below`.

Default: `auto`

**clip** `auto` or `bool`

See `std.block.clip`.

Default: `auto`

**sticky** `auto` or `bool`

See `std.block.sticky`.

Default: `auto`

**..args** `arguments`

Accepts at most 1 positional argument: `[body]`, the content of the minipage. See `std.block.body`.

### `deixis-minipage()`

Creates an isolated layout environment for notes, utilizing `margin` instead of `inset/outset`.

This is just a convenience wrapper around `#deixis-block`. It forward calls to `#deixis-block` with `inset: margin` and `outset: 0pt` by default.

See `#deixis-block` for details about parameters.

## Parameters

```
deixis-minipage(  
  id,  
  thread,  
  sync-counters-with,  
  width,  
  height,  
  breakable,  
  fill,  
  stroke,  
  radius,  
  margin: auto length dictionary ,  
  spacing,  
  above,  
  below,  
  clip,  
  sticky,  
  ..args  
) -> content
```

**margin** auto or length or dictionary

External margin pushing the block outward from surrounding content.

Default: auto

## 2.7. Utilities

### 2.7.1. Pin

#### `deixis-set-pin-pattern()`

Sets the global prefix and postfix used by `deixis-attach()` regex engine.

##### Parameters

```
deixis-set-pin-pattern(  
  prefix: str,  
  postfix: str  
)
```

**prefix**    `str`

The string prefix to use for regex matching.

Default: `auto`

**postfix**    `str`

The string postfix to use for regex matching.

Default: `auto`

#### `deixis-pin()`

Manually drops an invisible, named geometric anchor (a "`pin`") into the document flow.

Pins are the coordinate anchors that `deixis-region-mark()` uses to draw its highlights. They can also be used as anchor coordinates for `deixis-inset-note-body()` and `deixis-margin-note-body()`.

##### Parameters

```
deixis-pin(  
  name: str,  
  padding: str | length | dictionary ,  
  extra-data: dictionary  
) -> content
```

**name**    `str`

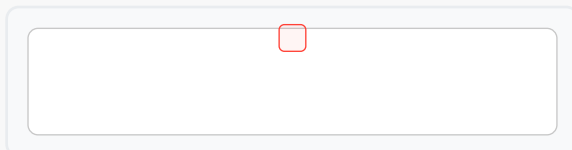
The unique identifier for this pin.

**padding**    `str` or `length` or `dictionary`

The mathematical padding around the pin.

- Accepts `length` or `dictionary`.

```
#place(center + horizon, deixis-  
pin("test-padding", padding: 5pt))  
#deixis-region-mark(pins: ("test-  
padding",))
```



- If "`text`", the height of the current font size is calculated.

```
The library was filled with the #deixis-
pin("susurrus-l", padding:
"text")susurrus#deixis-pin("susurrus-r",
padding: "text") of turning pages.
#deixis-region-mark(pins: ("susurrus-l",
"susurrus-r"))
```

The library was filled with the **susurrus** of turning pages.

Default: "text"

**extra-data** dictionary

Additional metadata to attach to the pin payload.

Default: ( : )

## deixis-attach()

A wrapper that attaches pins to a block of content.

This function does two things:

1. It scans the body text for your pin prefix (e.g., deixispinTop) and converts them into physical inline pins.
2. It can place absolute overlay pins relative to the bounding box of the body (e.g., placing a pin exactly at dx: 100%, dy: 100%).

### Parameters

```
deixis-attach(
  body: content,
  pins: dictionary,
  prefix: auto str none,
  postfix: auto str none,
  padding: auto str length dictionary,
  inline: bool
) -> content
```

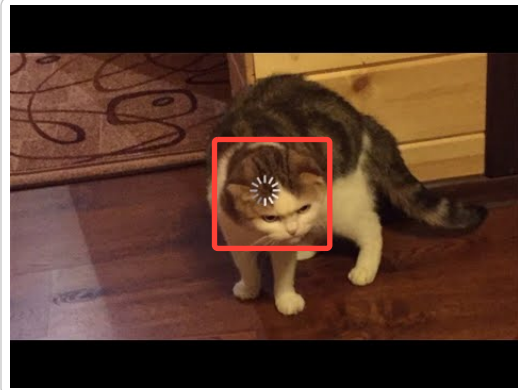
**body** content

The content block to wrap and scan.

**pins** dictionary

A dictionary mapping pin names to overlay coordinates and padding. Example: ("my-pin": (dx: 100%, dy: 50%, padding: 2pt)).

```
#deixis-attach(
  pins: (
    cat-tl: (dx: 40%, dy: 35%, padding:
(left: -5pt, top: -5pt)),
    cat-br: (dx: 60%, dy: 60%, padding:
(right: 5pt, bottom: 5pt)),
  ))[
  #image("../assets/loading-cat.jpg")
]
#deixis-region-mark(
  pins: ("cat-tl", "cat-br"),
  stroke: red + 2pt,
)
```



Default: (:)

**prefix** auto or str or none

The regex prefix to search for. If auto, it uses the global state. Set to none to disable regex scanning.

### **i** Info

Pins attached with regex scan will have padding: "text" by default. Use `deixis-attach.padding` for customization.

Usage:

```
#deixis-attach(prefix: "mypin")[
  ``typst
  #import "@preview/deixis:0.1.0": *

  mypincode1#deixis-
attachmypincode2(postfix: "deixis")[
  deixispin1deixisThis line will be
pinned.deixispin2deixis
]
  // Now one can use pins "p1" and "p2"
  ``
]
#deixis-region-mark(pins: ("code1",
"code2"), fill: red.transparentize(95%))
```

Usage:

```
#import "@preview/deixis:0.1.0": *

#deixis-attach(postfix: "deixis")[
  deixispin1deixisThis line will be
pinned.deixispin2deixis
]
  // Now one can use pins "p1" and "p2"
```

### **🔥** Tip

Oftentimes, if prefix alone is not enough, you can also define a postfix.

The name of the pin attached with regex scan should also not contain any special character, depending on the raw language.

Default: auto

**postfix** auto or str or none



The regex postfix to search for. If `auto`, it uses the global state.

```
Python hello world program:
#deixis-attach(
  prefix: "mypin",
  postfix: "mypin")[
  ``python

mypincode3mypinprintmypincode4mypin("Hello
world!")
  ``
]
#deixis-region-mark(pins: ("code3",
"code4"), fill: red.transparentize(95%))
```

Python hello world program:

```
print("Hello world!")
```

Default: `auto`

**padding** `auto` or `str` or `length` or `dictionary`

Padding for pins replaced with `deixis-attach.prefix`. It has no effect on pins defined with `deixis-attach.pins`.

```
Pythagorean theorem:

#align(center)[
#deixis-attach(
  prefix: "mypin",
  padding: ( // dict of pin names
    eq1: (top: 1.2em, bottom: 0.5em,
left: 1em),
    eq2: (top: 1.2em, bottom: 0.5em,
right: 1em),
  ),
)[
  mypineq1$a^2 + b^2 = c^2$mypineq2
]
]
#deixis-region-mark(pins: ("eq1", "eq2"),
stroke: orange, fill:
orange.transparentize(95%))
```

Pythagorean theorem:

$$a^2 + b^2 = c^2$$

Default: `auto`

**inline** `bool`

If `true`, the wrapper uses an inline box instead of a block.

```
#lorem(7)
#deixis-attach(
  pins: (tl1: (dx: 0%, dy: 0%), br1: (dx:
100%, dy: 100%)),
  inline: false)[
  Block-level attached content.
]
#deixis-region-mark(pins: ("tl1", "br1"),
stroke: teal)

#lorem(7)
#deixis-attach(
  pins: (tl2: (dx: 0%, dy: 0%), br2: (dx:
100%, dy: 100%)),
  inline: true)[
  Inline attached content.
]
#deixis-region-mark(pins: ("tl2", "br2"),
stroke: green)
```

Lorem ipsum dolor sit amet, consectetur  
adipiscing.

Block-level attached content

Lorem ipsum dolor sit amet, consectetur  
adipiscing. Inline attached content

Default: `false`

### **deixis-pin-prefix-state** `str`

The default prefix used in `deixis-attach.prefix` when `prefix: auto`. Defaults to `"deixispin"` and can be set using `deixis-set-pin-pattern()`.

#### **Warning**

While counter-intuitive, prefix should not contain special characters like `".~!@#$$%^&*()-_"`, if you want it to work on raw code.

### **deixis-pin-postfix-state** `str`

The default postfix used in `deixis-attach.postfix` when `postfix: auto`. Defaults to `""` (empty string) and can be set using `deixis-set-pin-pattern()`.

#### **Warning**

Similar to `deixis-pin-prefix-state`, postfix should not contain special characters like `".~!@#$$%^&*()-_"`.

### 2.7.2. Counter

#### deixis-note-counter()

Retrieves the current integer count of notes for a specific target block and series.

Because deixis relies on a custom state engine to track scoped notes, you cannot use Typst's native `counter()` function to read note counts. This function provides direct access to the internal engine.

Note: Because it queries the document state, this function must be called within a `context` block.

```
#context deixis-note-counter()
```

34

#### Parameters

```
deixis-note-counter(  
  target: int | label | str,  
  series: str  
) -> int
```

**target** `int` or `label` or `str`

The specific block or minipage to query. Defaults to `0` (the current block). Accepts a label/string ID, or a negative integer to query relative parent blocks.

Default: `0`

**series** `str`

The specific note series to query.

Default: `"default"`

#### deixis-update-note-counter()

Set or update the note counter for a specific target block and series.

This is highly useful for manually resetting note counts at the start of a new chapter or section if you are not using an automatic minipage wrapper. It modifies the internal `deixis` state dictionary.

```
#deixis-update-note-counter(0)  
#context deixis-note-counter()
```

0

#### Parameters

```
deixis-update-note-counter(  
  value: int | function,  
  target: int | label | str,  
  series: str  
) -> content
```

**value** `int` or `function`

The new integer value for the counter. Alternatively, pass a function `(current-count) => int` to modify the count relative to its current value.

**target** `int` or `label` or `str`

The specific block or minipage whose counter should be updated. Defaults to `0` (the current block). Accepts a label/string ID, or a negative integer for relative parent blocks.

Default: `0`

**series** `str`

The specific note series to update.

Default: `"default"`

### 2.7.3. Cross-reference

#### deixis-ref()

A styled, clickable cross-reference to a specific note.

##### Parameters

```
deixis-ref(  
    ..targets: arguments ,  
    supplement: auto content none ,  
    marker-style: auto function ,  
    celibate-marker: str  
) -> content
```

**..targets** arguments

The positional target `label` (e.g., `<my-note>`) to reference.

Let's create some test notes  
`#deixis-margin-note(label: <note-a>)[Test note A.]`  
`#deixis-margin-note(label: <note-b>)[Test note B.]`  
`#deixis-margin-note(label: <note-c>)[Test note C.]`  
  
Single reference: `#deixis-ref(<note-a>)`

Let's create some test notes<sup>123</sup>  
Single reference: <sup>1</sup>

- Multiple consecutive references will be collapsed automatically.

Multiple reference: `#deixis-ref(<note-a>, <note-b>, <note-c>)`

Multiple reference: <sup>1-3</sup>

- The last positional argument, if of type `content` or `str`, will be interpreted as `deixis-ref.supplement`.

`#deixis-ref(<note-b>)[Comment]`

Comment <sup>2</sup>

**supplement** auto or content or none

The prefix to display before the number (e.g., [Note]). Set to `none` for just the number.

`#deixis-ref(<note-c>, supplement: [Todo])`

Todo <sup>3</sup>

Default: auto

**marker-style** auto or function

A custom styling function (string) => content for the reference number. If `auto`, it inherits the note's original marker style.

Default: auto

<sup>2</sup>Test note B.

<sup>1</sup>Test note A.  
<sup>3</sup>Test note C.

**celibate-marker** `str`

The marker displayed for celibate notes.

Default: "?"

### **deixis-show-refs()**

A show rule to automatically convert @note-label to #**deixis-ref**(<note-label>).

**#show:** `deixis-show-refs`

#### **Parameters**

`deixis-show-refs`(body) -> `content`

## 2.7.4. Note Outline

### deixis-note-outline()

A dynamically styled outline (Table of Contents) for your annotations.

The outline extracts the content from note bodies and clips it to a single line. Each entry features a dynamic visual badge that inherits the styling of the specific note it represents. They act as clickable hyperlinks, returning the reader to the mark's location.

```
#deixis-inset-note(dx: 10pt, dy: 0pt,  
stroke: red)[This is a note!]  
  
#deixis-inset-note(dx: 10pt, dy: 0pt,  
stroke: green)[This is another note!]  
  
#deixis-inset-note(dx: 10pt, dy: 0pt,  
stroke: blue)[This is yet another note!]  
// `target: 0` restricts the search to  
notes inside this example block.  
#deixis-note-outline(target: 0)
```

- 1 <sup>1</sup>This is a note!
- 2 <sup>2</sup>This is another note!
- 3 <sup>3</sup>This is yet another note!

### List of Notes

|   |                           |    |
|---|---------------------------|----|
| 1 | This is a note!           | 87 |
| 2 | This is another note!     | 87 |
| 3 | This is yet another note! | 87 |

### Parameters

```
deixis-note-outline(  
  title: content or none ,  
  heading-level: int ,  
  target: auto label str int ,  
  thread: auto str ,  
  include-types: auto array ,  
  exclude-types: array ,  
  series: auto str array ,  
  include-celibates: bool str ,  
  after: none location label str ,  
  before: none location label str ,  
  group-by: none str function ,  
  fill: none content ,  
  gap: auto length  
) -> content
```

**title**    content or none

The heading content to display above the outline. Set to `none` to disable the heading.

```
#deixis-note-outline(target: 0, title:  
[Liste des Notes])
```

### Liste des Notes

Default: [List of Notes]

**heading-level**    int

The heading level for the outline's main title. Any dynamically generated group sub-headings will be rendered at heading-level + 1.

```
#deixis-note-outline(target: 0, heading-level: 3)
```

## List of Notes

Default: 1

**target** `auto` or `label` or `str` or `int`

Filters the outline to notes belonging to a specific target block or minipage.

Default: `auto`

**thread** `auto` or `str`

Filters the outline to only show notes belonging to a specific thread of minipages.

Default: `auto`

**include-types** `auto` or `array`

A list of note types to explicitly include (e.g., ("`margin-note`", "`endnote`"). If `auto`, includes all types.

Default: `auto`

**exclude-types** `array`

A list of note types to explicitly exclude.

Default: `()`

**series** `auto` or `str` or `array`

Filters the outline to notes belonging to a specific series or array of series.

Default: `auto`

**include-celibates** `bool` or `str`

Determines if decoupled, standalone marks or bodies should be included. Choices: `bool` | "`both`" | "`none`" | "`mark`" | "`body`".

Default: `false`

**after** `none` or `location` or `label` or `str`

Only includes notes that appear **after** this location, label, or `<deixis-pin>` name.

Default: `none`

**before** `none` or `location` or `label` or `str`

Only includes notes that appear **before** this location, label, or `<deixis-pin>` name.



Default: **none**

**group-by** **none** or **str** or **function**

Categorizes the notes under sub-headings. Accepts **"type"**, **"series"**, or a custom **function(note) => string**.

```
#deixis-footnote(stroke: red)[This is a footnote!]  
#deixis-margin-note(stroke: green)[This is a margin note!]  
#deixis-inset-note(dx: 10pt, dy: 0pt, stroke: blue)[This is an inset note!]  
  
#deixis-note-outline(target: 0, group-by: "type")
```

|                                  |                                     |                                     |
|----------------------------------|-------------------------------------|-------------------------------------|
| 123                              | <sup>3</sup> This is an inset note! | <sup>2</sup> This is a margin note! |
| <b>List of Notes</b>             |                                     |                                     |
| <b>Footnotes</b>                 |                                     |                                     |
| <sup>1</sup>                     | This is a footnote!                 | 89                                  |
| <b>Margin Notes</b>              |                                     |                                     |
| <sup>2</sup>                     | This is a margin note!              | 89                                  |
| <b>Inset Notes</b>               |                                     |                                     |
| <sup>3</sup>                     | This is an inset note!              | 89                                  |
| <hr/>                            |                                     |                                     |
| <sup>1</sup> This is a footnote! |                                     |                                     |

Default: **none**

**fill** **none** or **content**

Fills the space between the excerpt and the page number. Commonly set to **repeat [ . ]** for standard TOC styling.

```
#deixis-inset-note(dx: 10pt, dy: 0pt, stroke: red)[This is a note!]  
  
#deixis-inset-note(dx: 10pt, dy: 0pt, stroke: green)[This is another note!]  
  
#deixis-inset-note(dx: 10pt, dy: 0pt, stroke: blue)[This is yet another note!]  
  
#deixis-note-outline(target: 0, fill: repeat[.])
```

|                      |                                        |    |
|----------------------|----------------------------------------|----|
| 1                    | <sup>1</sup> This is a note!           |    |
| 2                    | <sup>2</sup> This is another note!     |    |
| 3                    | <sup>3</sup> This is yet another note! |    |
| <b>List of Notes</b> |                                        |    |
| <sup>1</sup>         | This is a note! .....                  | 89 |
| <sup>2</sup>         | This is another note! .....            | 89 |
| <sup>3</sup>         | This is yet another note! .....        | 89 |

Default: **none**

**gap** **auto** or **length**

The vertical spacing between individual note rows in the outline.

Default: **auto**

## 2.7.5. Others

### deixis-page-notes()

Fetches all rendered deixis notes that exist on or visually intersect the current page.

This is an useful introspection tool and can be used, for example, to set page headings based on what endnote is being displayed.

#### **i** Info

Since it relies on spatial queries, it must be called within a **context** block.

```
#let adaptive-header = context {
  let notes = deixis-page-notes(note-type: "endnote")

  if notes.len() == 0 {
    align(center)[_My Book Title_]
  } else {
    let first-note = notes.first()
    let last-note = notes.last()
    // Helper to find the page number of the mark in the main text
    let _find-mark-page(lbl) = {
      let elems = query(selector(lbl))
      if elems.len() > 0 { elems.first().location().page() } else { 0 }
    }
    let p-start = _find-mark-page(first-note.mark-lbl)
    let p-end = _find-mark-page(last-note.mark-lbl)
    // pages
    let left-text = if p-start == p-end { [_Notes for page #p-start _] } else { [_Notes
for pages #p-start -- #p-end _] }
    // note numbers
    let right-text = if notes.len() == 1 { [_Note #first-note.marker-str _] } else
{ [_Notes #first-note.marker-str -- #last-note.marker-str _] }
    grid(
      columns: (1fr, 1fr),
      align(left)[#left-text],
      align(right)[#right-text]
    )
  }
}

#adaptive-header
// Real usage: (cannot be used in example environment)
// #set page(header: adaptive-header)

We can do that, Rik
#deixis-endnote[Test note 1 #lorem(10)]
#deixis-endnote[Test note 2 #lorem(10)]
#deixis-endnote[Test note 3 #lorem(10)].

#deixis-print-endnotes()
```

*Notes for page 91*

*Notes 1 – 3*

We can do that, Rik<sup>123</sup>.

- 1 Test note 1 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do.
- 2 Test note 2 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do.
- 3 Test note 3 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do.

### Parameters

`deixis-page-notes`(note-type) -> `array`

### `deixis-place-anchored()`

Places content relative to a specific anchor mark given its `internal-id` or a pin given its name.

### Parameters

```
deixis-place-anchored(  
  body,  
  dx,  
  dy,  
  anchor,  
  internal-id,  
  pin  
)
```

### `deixis-absolute-place()`

Places content at a specific exact absolute coordinate on the page ignoring page margins.

### Parameters

`deixis-absolute-place`(.args)

## 3. Limitations and Conclusion

### 3.1. Known Issues

If you encounter any bug, please report at <https://github.com/inspiros/typst-deixis>.

This is just a non-exhaustive list of known limitations that might not have a fix.

- Page-level footnotes are currently dependent on `std.footnote` internally, limiting their customizability.
- Occasionally, if put inside the demo environment in this manual, foreground overlayed notes like footnotes and margin notes may have their mark → body link broken, pointing to the beginning of the document when clicked.
- Similarly, a relative-placed `#deixis-inset-note` sometimes have wrongly calculated body coordinates, resulting in a link connecting to the border of the page.

```
#deixis-inset-note(  
  stroke: red,  
  fill: red.transparentize(95%),  
  link: "straight-line",  
  link-marks: "both",  
  dx: 4em, dy: 3em,  
  layer: "flow", // not affecting  
  "foreground"  
) [Note mark.1] [Note body.]
```

Note mark.<sup>1</sup>

<sup>1</sup>Note body.

### 3.2. Conclusion



#### Conclusion

The whole package is experimental, a proof of concept. Do not expect much, as there are always limits to what a hacky Typst package can do.

Plus, having a strong opinion on using native features whenever possible, I hope future Typst versions will provide a more powerful noting mechanism.